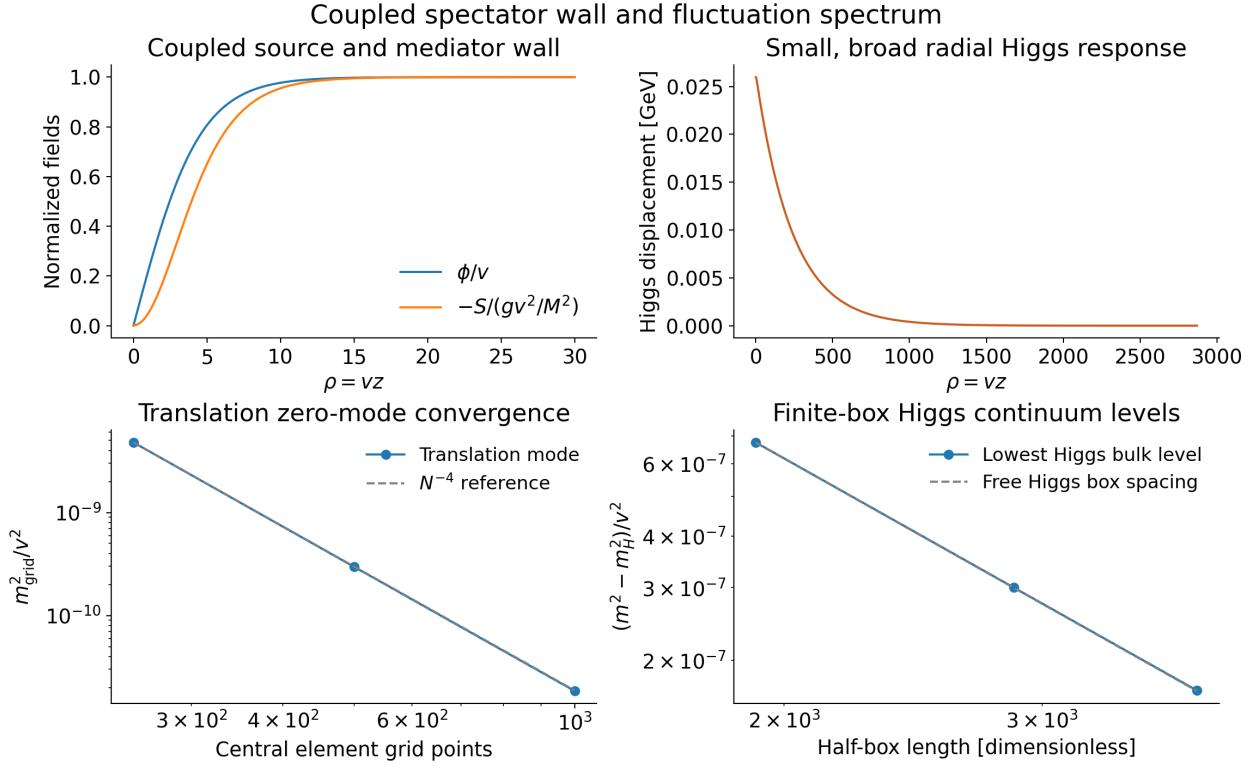


Coupled wall stability, exact thermal vacua and polynomial flavor bounds

This advance builds the canonical three-field spectator wall, calculates its finite-box fluctuation spectrum in both reflection sectors, and classifies the supplied mean-field thermal potential exactly. The Higgs transition occurs at 140.24158166969536 GeV, below the CP transition at 59999.960002080275 GeV. The radial Higgs at the unbiased wall center is 246.02597579488483 GeV. Its low propagation threshold, approximately 125.44 GeV, is essential: the 11.62 TeV scalar reference shape mode lies in an open Higgs channel. It cannot be called an isolated coupled bound state without a resonance calculation. The chosen cosmological inputs, earlier conditional gravitational-wave estimates and nonet flavor limitations remain as stated in the retained chapters below.



The scalar limit and the coupled fluctuation problem. Write $\rho=v z$, $u=\phi/v$, $y=S/v$, $\eta=h_{\text{radial}}/v$, $\mu=M/v$, $\alpha=g/v$ and $\eta_0=v_H/v$. The potential in units of v to the fourth is $\lambda(u^2-1)^2/4 + \mu^2(y+\alpha u^2/\mu^2)^2/2 + \lambda_H[\eta^2-\eta_0^2+(\kappa/\lambda_H)(u^2-1)]^2/4$. The canonical gradient energy is one half the sum of the three derivative squares. There is no bias in the static boundary problem: the late biased vacua are not degenerate and do not support the same stationary wall. All boundary conditions are imposed on a half-line using u odd and y, η even. At the origin $u=0$ and $y'=\eta'=0$; at the outer boundary $u=1$, $y=-\alpha/\mu^2$ and $\eta=\eta_0$.

The coupled Euler equations are $u''=\lambda u(u^2-1)+2\alpha u r_S+\kappa u r_H$, $y''=\mu^2 r_S$, and $\eta''=\lambda_H \eta r_H$, where $r_S=y+\alpha u^2/\mu^2$ and $r_H=\eta^2-\eta_0^2+(\kappa/\lambda_H)(u^2-1)$. The BVP uses a graded mesh because the source inverse width is $\sqrt{\lambda/2}$ in ρ units, while the Higgs inverse length is $\sqrt{2\lambda_H}\eta_0$. The reference half-box extends to $\rho=2869.9922215344004$, or twelve Higgs decay lengths. The tension is approximately $8.04984793435 \times 10^{12}$ GeV cubed and lies in the prior exact three-field tension sandwich. Its computed first-integral error is below ten to the minus twelve. Numerical quadrature error is not a rigorous total BVP error bound, and the tiny portal correction to the enormous total tension is not assigned a certified uncertainty here.

A separate linear-response calculation solves the Higgs equation around the source kink with the massive one-dimensional Green function. It predicts the central displacement $\kappa\eta_0/m_H$ times the integral from zero to infinity of $\exp(-m_H\rho)\text{sech}(\sqrt{\lambda/2}\rho)^2$, multiplied by v . The result is 0.02597445705679549 GeV, compared with the nonlinear three-field value 0.02597579488483 GeV. Their relative difference is about 5.2 times ten to

the minus five. This comparison checks the broad Higgs tail through an independent approximation; it does not replace the nonlinear solution.

For the isolated scalar kink, let $x=k z$ with $k=v*\sqrt{\lambda/2}$. Its Hessian operator is k squared times $[-d^2/dx^2+4-6 \operatorname{sech}(x)^2]$. Exactly, this equals k squared times $(-d/dx+2 \tanh x)(d/dx+2 \tanh x)$. The translation eigenfunction $\operatorname{sech}(x)^2$ has eigenvalue zero; the shape eigenfunction $\operatorname{sech}(x)*\tanh(x)$ has eigenvalue $3 k$ squared; and the continuum begins at $4 k$ squared. For $v=30000$ GeV and $\lambda=0.1$, the shape mass is 11618.95003862225 GeV and the source continuum threshold is approximately 13416.4 GeV. The factorization proves nonnegative scalar fluctuations without a sampled eigenvalue calculation. This statement is about the scalar reference, not a theorem about all coupled fluctuations.

The coupled Hessian is evaluated by differentiating the displayed polynomial potential, and is checked independently by symbolic differentiation in the tests. On the half-box, the translation sector has δu even and $\delta y, \delta \eta$ odd, so its left boundary conditions are Neumann, Dirichlet, Dirichlet. The complementary sector has Dirichlet, Neumann, Neumann. Both use fixed outer fields. Quadratic conforming finite elements and five-point Gauss potential integration give sparse symmetric stiffness and consistent mass matrices. A floating banded Cholesky factorization independently tests positivity of the complete discretized stiffness matrix; shift-invert eigenvectors identify its low levels. The calculation includes canonical radial Higgs and spectator fluctuations only. Gauge fields, angular Higgs directions, fermionic fluctuations and thermal dissipative modes are outside this static operator.

All ten matrices from three central meshes and two additional box lengths are positive in floating arithmetic. With central grids of 250, 500 and 1000 points, the near-zero translation eigenvalue divided by v squared is approximately 4.79 times ten to the minus nine, 2.97 times ten to the minus ten and 1.85 times ten to the minus eleven. The decrease by roughly sixteen per doubling is the expected fourth-order eigenvalue convergence of quadratic elements. Translation overlap exceeds 0.99999999. These small positive eigenvalues are discretization offsets, not predicted particle masses. Normalized backward eigen residuals are recorded. Positivity and mesh convergence give numerical evidence of stability; they do not certify the infinite-domain coupled spectrum or exclude every unresolved negative mode.

At the homogeneous vacuum the three continuum thresholds correspond to masses about 125.43588 GeV, 13416.38098 GeV and 300000.60120 GeV. The first is the radial Higgs channel. The lowest nontranslation levels move toward its threshold when the half-box is enlarged from eight to sixteen Higgs decay lengths. A portal-free control reproduces the independent free-Higgs box spacings $m_h^2+(\pi/(2L))^2$ and $m_h^2+(\pi/L)^2$ in the complementary parity sectors. Thus low box levels are bulk propagation states, not a collection of new wall particles. The scalar shape excitation can couple to this continuum when κ is nonzero. Its resonance position, width and scattering phase have not been calculated.

Exact homogeneous thermal selection. Add one half $c*T^2*\phi^2$ and one half $c_H*T^2*h_{\text{radial}}^2$, with $c=0.025000033333333335$ and $c_H=0.4$ declared inputs. Minimize S exactly. For zero bias introduce $x=u^2 \geq 0$ and $w=\eta^2 \geq 0$. The remaining potential is a quadratic in x, w . Twice its amplitude Hessian is $G=[[\lambda+\kappa^2/\lambda_H, \kappa], [\kappa, \lambda_H]]$, with positive leading minor and exact determinant $\lambda*\lambda_H=0.013$. It is strictly convex and coercive. Its unique global amplitude minimum is therefore given by the exact rational KKT conditions $G*(x, w)=\text{rhs}$ on positive coordinates and nonnegative outward derivatives on zero coordinates. The vector rhs is $(\lambda+\kappa*\eta_0^2+\kappa^2/\lambda_H-c*(T/v)^2, \lambda_H*\eta_0^2+\kappa-c_H*(T/v)^2)$. The implementation checks all four faces of the nonnegative amplitude quadrant and retains boundary coincidences as one physical amplitude.

At zero temperature, $x=1$, $w=(246/30000)^2$ and $y=-1/1000$ exactly. The two signs of ϕ give the two CP-related global components, modulo Higgs gauge orientations. At high temperature $x=w=0$. Between the CP and Higgs transitions the source is broken while the Higgs is symmetric; below the Higgs transition both are broken. The exact squared transition temperatures are stored as rational numbers, avoiding a floating classification at a bifurcation. Their displayed square roots are leading mean-field temperatures of the supplied thermal action, not gauge-resummed physical transition predictions. The first CP transition is outside the chosen late bias onset, so the unbiased classification applies there.

The chosen bias is $h(T)=h_0*\max[0, 1-(T/T_{\text{onset}})^2]$, $T_{\text{onset}}=3000$ GeV. For active positive bias $b=h/v$ cubed, configurations at opposite source signs and the same amplitude differ by $2*b*|u|$, so the positive sign wins globally. On that sign, the extra potential term is $-b*\sqrt{x}$, a convex function on $x \geq 0$. Together with the strictly convex

amplitude quadratic this proves a unique global amplitude minimum with $x > 0$. This is an exact global selection result for the declared spectator thermal model. It is not a derivation of the bias, a solution of the nonet flavor vacuum, or a cosmological constant prediction.

All stationary branches remain useful, including the false vacuum and the saddles at $h_{\text{radial}} = 0$. On the Higgs boundary the source cubic is $(\lambda + \kappa^2/\lambda_H)u^3 - [\lambda + \kappa^2\eta_0^2 + \kappa^2/\lambda_H - c^2/(T/v)^2]u - b = 0$. On the Higgs-broken branch it is $\lambda u^3 - [\lambda - (c\kappa/\lambda_H)(T/v)^2]u - b = 0$, with $w = \eta_0^2 + (\kappa/\lambda_H)(1 - u^2) - c^2/(T/v)^2 > 0$. The boundary branch remains stationary even when its radial Higgs curvature is negative and is explicitly retained as a saddle. Ten temperature snapshots record all distinct real branches, their rational root enclosures, exact curvature signs and global selection. At 30 MeV the full radial census contains six stationary branches: two local minima, their source barrier, and three Higgs-boundary saddles. Only the positive-source local minimum is global under the active bias.

Branch preservation and the OpenAI Math connection. The module reuses Perfectpower rational polynomial, gcd and Sturm-chain primitives. Repeated roots are isolated through the squarefree part while the multiplicity gcd is kept in the receipt; exact dyadic roots are deflated before restarting isolation so endpoint conventions cannot delete them. A gcd between the defining polynomial and a test polynomial recognizes exact zero at an algebraic branch; interval refinement decides the remaining signs. The fixture $(u-1)^2(u+2)$ retains its double root, zero curvature and vanishing $u-1$ denominator. This is a mathematical fixture, not a cosmological benchmark spinodal. It applies the branch-preserving idea from OpenAI Math family 142, whose finite-field section describes simultaneous quotient execution and gcd splitting. Family 141 motivates real algebraic labels and universal constrained selection; here the particular potential permits the much simpler exact convex-amplitude proof. Families 262 and 375 motivate spectral and quartic-interface analysis, while the scalar factorization and the coupled numerical calculation are derived directly here. No headline conjecture claim from those manuscripts is an input to this result.

Primary source references. OpenAI Math, github.com/openai/math, family 142: Deterministic Polynomial Factorization over Prime Fields (October 4, 2026), section Finite-field computations without known roots. Family 141: Existential-universal real sentences in the counting hierarchy (October 4, 2026), section From two real blocks to two existential tests. Family 262: Equality cases in the sharp one-dimensional matrix Lieb-Thirring inequality (October 5, 2026), section An action estimate that retains the equality defects. Family 375: A positive resolution of De Giorgi's conjecture in dimension eight (September 26, 2026), section Two classical estimates for the quartic equation. The catalogue was inspected at tree `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. These are source inspirations with their own stated hypotheses, not imported guarantees for our flavor theory.

Validation, reproduction and coordination. Sixteen new wall and thermal-branch tests, thirteen adjacent dimensionful-wall tests and forty-six adjacent flavor tests passed, for seventy-five focused tests. The scientific receipt regenerates byte for byte with SHA256 `53dba2e4f0f07a2bd22ce3efb50bdcd280df35ca61039f89b927d62ff333f0f`. The full repository regression, incoming singlet tests and Lean build were not rerun in this wall-specific round. Run `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 python python/develop_wall_stability.py`, followed by `python scripts/plot_wall_stability.py`. Run the focused wall tests with `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 python -m unittest discover -s python/tests -p test_wall_stability.py`. The receipt consumes the existing dimensionful benchmark by hash and uses a deterministic eigenvector start. The previous gravity-wave templates and 162-case cosmological screen are retained; this advance does not calibrate network formation or annihilation. New module and test paths are specific to spectator walls. The concurrent singlet mass-correlation and protected-operator release `f0db543fb52f02c2b9453083efe42cfc8d586d57` is preserved and does not overlap this implementation. The earlier chapters follow as retained derivations and records, with the exact spectator thermal selection and three-field static calculation above superseding their missing three-field calculations.

This release supplies the dimensionful wall quantities that the earlier conditional interface deliberately left unspecified. It constructs a concrete spectator CP-wall benchmark on the repository's Gaussian positive-square model, solves its two-field static wall, and calculates a mean-field transition temperature, an explicit late-bias annihilation screen, radiation-era gravitational-wave spectra and a 162-point network sensitivity scan. It also supplies a weak CP-even Higgs-bath portal with a calculable scalar decay channel. The point passes the declared wall-network, radiation-budget and timing screens throughout that chosen scan. It is a model benchmark, not a cosmologically validated quark-flavor

benchmark: the retained nonet CP-breaking branch still has a lower CP-conserving competitor and uncontrolled quantum feedback, and no choice of units changes those facts.

The declared source scale is $v=30,000$ GeV, with $\lambda=0.1$, a canonical Gaussian mediator of mass $M=300,000$ GeV and current coupling $g=3,000$ GeV. The unbiased potential is $\lambda(\phi^2-v^2)^2/4 + M^2(S+g\phi^2/M)^2/2$. Both terms are nonnegative, and their simultaneous zeros prove the only physical source minima are $\phi=\pm v$ and $S=-g v^2/M$. CP sends ϕ to $-\phi$ and leaves S invariant. These inputs are chosen scales and couplings. They are not derived from observed quark masses, a golden CKM relation or a fundamental prediction for the flavor scale.

The numerical static-wall boundary problem uses $\rho=vz$, $u=\phi/v$ and $y=S/v$. With $\alpha=g/v$ and $\mu=M/v$, its equations are $u''=\lambda u(u^2-1)+2\alpha u(y+\alpha u^2/\mu^2)$, and $y''=\mu^2(y+\alpha u^2/\mu^2)$. The boundary values are $u=\pm 1$ and $y=-\alpha/\mu^2$. An adaptive two-field boundary solver gives $\sigma=8.04984793435$ times 10 to the twelfth GeV cubed. Expanding the domain and tightening the residual tolerance changes the result by 2.04 times 10 to the minus ten relatively. The conserved static first integral and endpoint residuals are also checked. This is a numerical differential-equation result, not a formally certified integrator.

A much sharper analytic tension bound is available than the earlier Cauchy path bound. Every full profile has energy at least that of its projected canonical source profile, so $\sigma \geq \sigma_0 = 2\sqrt{2\lambda} v^3/3$. Conversely lift the exact source kink $\phi=v \tanh[v \sqrt{\lambda/2}z]$ onto $S=-g \phi^2/M$. Its Gaussian potential square vanishes, and the extra heavy kinetic energy integrates exactly to $\sigma_0 \times 2 g^2 v^2/(5 M^4)$. Thus $\sigma_0 \leq \sigma \leq \sigma_0[1+2 g^2 v^2/(5 M^4)]$. Here the relative trial excess is exactly $1/2500000$. The resulting tree-action interval is 8.04984471900 times 10 to the twelfth to 8.04984793894 times 10 to the twelfth GeV cubed. The numerical solution lies inside it. These exact bounds concern the supplied tree potential; they do not include an unspecified loop-corrected thermal action.

The spectator bath completion adds $\lambda_H/4$ times $[2 H^\dagger H - v_H^2 + (\kappa/\lambda_H)(\phi^2 - v^2)]^2$, with $\kappa=10$ to the minus seven, $\lambda_H=0.13$ and $v_H=246$ GeV. This remains a positive square and vanishes at both source minima with $2 H^\dagger H = v_H^2$. Holding H at its vacuum in the lifted source-kink trial adds a relative potential cost $\kappa^2/(2 \lambda_H \lambda)$, exactly $1/260000000000$. The resulting three-field tension upper bound is only about 3.10 GeV cubed above the two-field trial upper bound. The two-field numerical profile does not purport to solve this third-field boundary problem, but the enlarged analytic sandwich remains valid. The leading ϕ -to-two-Higgs decay width is $\kappa^2 v^2/(8 \pi m_\phi)$ times $\sqrt{1-4 m_h^2/m_\phi^2}$. It is 2.669 times 10 to the minus eleven GeV, with lifetime 2.47 times 10 to the minus fourteen seconds, so this example does not simply leave all wall energy in a stable heavy scalar. A complete relic and entropy history still needs Boltzmann evolution.

The specified mean-field thermal action adds $c T^2 \phi^2/2$, with $c=\lambda/4+\kappa/3$, and $c_H T^2$ times $2 H^\dagger H/2$, with $c_H=0.4$. The latter is a supplied thermal-bath coefficient. At the high-temperature CP transition the Higgs curvature is positive, about 1.44 times 10 to the ninth GeV squared, so its minimum is $H=0$. The continuous CP transition then has $T_c^2 = [\lambda v^2 + \kappa v_H^2 + \kappa^2 v^2/\lambda_H]/c$. This gives $T_c=59,999.9600$ GeV, approximately 60 TeV. The source-only estimate differs by roughly two millionths of a GeV. The leading source transition is second order in this specified free energy, not a first-order bubble-nucleation signal. Gauge resummation, additional thermal operators and critical dynamics can change a physical transition; none is implicitly included in this mean-field number.

To permit actual disappearance of the walls, the benchmark declares an explicitly CP-odd late bias $-h(T) \phi$, with $h(T)=h_0 \max[0, 1-(T/3000 \text{ GeV})^2]$. It vanishes near T_c , so it does not replace the formation transition with an explicitly biased crossover. Its onset history is an input rather than an interaction mechanism derived by the matching session. The chosen $h_0=2.58436623$ times 10 to the minus thirteen GeV cubed targets reference annihilation at 30 MeV. That temperature is therefore a design target, not an independently predicted output. A maximum post-inflation temperature ten times T_c supplies the stated restoration history. Defect formation, domain populations and the origin of the late bias still require their own dynamics.

The tiny bias makes ordinary double-precision subtraction of the vacuum energies unreliable. The driver instead solves the cubic stationary equation with eighty decimal digits and evaluates the relaxed branch energies there. At the reference

annihilation temperature, both local basins survive; the gap is $1.55061974 \times 10^{-8}$ GeV to the fourth. Their zero-bias barrier is 2.025×10^{-16} GeV to the fourth. Low-temperature Higgs thermal feedback changes the source thermal coefficient by about 3.08×10^{-7} . The driver includes it by exact static minimization of the radial Higgs square, leaving $c_{\text{low}} = c_{\text{H}}/\lambda_{\text{H}}$. The leading tiny-bias formula $2 h(T) v \sqrt{(1 - c_{\text{low}} T^2)/(\lambda v^2)}$ agrees with the high-precision relaxed gap at the stated precision of the network calculation.

The radiation-era convention is explicit: $H = \sqrt{\pi^2 g_{\text{star}}/90} T^2/M_{\text{reduced}}$ with $M_{\text{reduced}} = 2.435 \times 10^{-18}$ GeV, and $t = 1/(2H)$. Define $\rho_{\text{wall}} = A \sigma/t = 2 A \sigma H$ and $t_{\text{ann}} = C_{\text{ann}} A \sigma/\Delta V$. The reference inputs $A=0.8$, $C_{\text{ann}}=3$ and $g_{\text{star}}=g_{\text{star},s}=10.75$ give T_{ann} approximately 30 MeV, $H_{\text{ann}} = 4.0131 \times 10^{-22}$ GeV, age 0.0008201 seconds and wall fraction 0.001804. The Higgs decay width exceeds this Hubble rate by more than ten billion. The zero-temperature true-vacuum constant is subtracted as a cosmological-constant input. Remaining false volume contributes $f_{\text{false}} \Delta V$ and is included in the energy screen. The declared caps are $f_{\text{false}} \leq 1/2$ at annihilation and $\leq 1/100$ at the end of emission; the latter requires actual network completion and is not computed by the static solver. A five-MeV completion deadline, maximum total defect fraction 0.1 and integrated GW radiation budget Ωh^2 below 10^{-6} are declared screening choices. They are not a complete observational likelihood or a fit to light-element abundance data.

The gravitational-wave calculation uses a conventional quadrupole efficiency and entropy-conserving redshift, with independently visible input coefficients. In terms of the reduced Planck mass, the emission peak fraction is $\epsilon_{\text{GW}} A^2 \sigma^2/[24 \pi M_{\text{reduced}}^4 H_{\text{GW}}^2]$. Its present amplitude is multiplied by $\Omega_{\text{rad}} h^2 (g_{\text{star}}/g_{\text{star},0})(g_{\text{star},s,0}/g_{\text{star},s})$ to the four-thirds. The peak frequency follows $f_0 = \beta H_{\text{GW}}/\hbar$ times $(T_0/T_{\text{GW}})(g_{\text{star},s,0}/g_{\text{star},s})$ to the one-third. Here β defines the literature cyclic-frequency convention, equivalently $k_{\text{em}} = 2 \pi \beta H_{\text{GW}}$. The reference inputs are $\epsilon_{\text{GW}}=0.7$ and $\beta=1$. Hiramatsu, Kawasaki and Saikawa, <https://arxiv.org/abs/1309.5001>, provide the simulation-based efficiency and radiation-era normalization used as the reference calibration.

In the reference scaling scenario, emission ends at $H_{\text{GW}}=H_{\text{ann}}$, with a peak at 3.4072×10^{-9} Hz and $\Omega_{\text{GW}} h^2 = 2.3613 \times 10^{-12}$. Its supplied spectrum rises as f^3 below the peak and falls as f^{-1} above it. A delayed-emission sensitivity case instead uses $H_{\text{GW}}=0.1 H_{\text{ann}}$, $\beta=2$, an initial post-peak slope minus one-half, a chosen second break at ten peak frequencies and subsequent slope minus 1.8. It peaks at 2.1549×10^{-9} Hz with amplitude 2.3613×10^{-10} . The factor of one hundred is the explicit H_{GW} to the minus two dependence under otherwise fixed inputs. This is a sensitivity scenario, not an imported lattice prediction for the benchmark potential. Notari, Rompineve and Torrenti, <https://arxiv.org/abs/2504.03636>, motivate treating late production and spectral shape as independent calibration questions.

The plotted curves are computed spectra under those declared assumptions. They are not a detection forecast, an explanation of pulsar-timing data or a universally calibrated domain-wall signal. Babichev and collaborators, <https://arxiv.org/abs/2504.07902>, find earlier bias-driven collapse and a different bias dependence in their simulated model, as well as a far-ultraviolet plateau. This prevents promoting the reference balance law to a general dynamics theorem. The release therefore also evaluates a generous plateau-cap radiation screen: below the peak use the causal f^3 shape, bound the spectrum by its peak up to the inverse-width cutoff, and allow one peak unit of integrated tail. Under that explicit envelope its log-frequency integral is bounded by $\Omega_{\text{peak}} h^2$ times $[\text{one third} + \log(f_{\text{width}}/f_{\text{peak}}) + \text{one}]$. This screen does not assume the two falling templates are the only allowed ultraviolet spectra.

The nuisance scan varies A over 0.6, 0.8 and 1, C_{ann} over 2, 3 and 5, efficiency over 0.3, 0.7 and 1.1, common radiation/entropy degrees over 10.75 and 17.25, and emission Hubble ratio over 0.1, 0.3 and 1. All 162 combinations pass the declared timing, conservative scaling-energy and plateau-cap radiation screens. Their peak frequencies span 1.435 to 5.278 nanohertz, and peak amplitudes span 1.216×10^{-13} to 2.516×10^{-9} in Ωh^2 . The lowest emission temperature is 5.84 MeV, the greatest conservative scaling wall fraction is 0.04699, the greatest wall-plus-false-volume fraction under the declared completion cap is 0.07049, and the largest plateau-cap integral is 1.554×10^{-7} . These are extrema of stated input choices, not confidence intervals; a different bias-to-annihilation calibration or thermal equation of state can move them.

The implementation is `python/perfectpower/dimensionful_walls.py`, with driver `python/develop_dimensionful_walls.py`, thirteen focused tests, the scientific receipt `receipts/flavor_cosmology/dimensionful_walls.json` and its plotted spectra. Run `PYTHONPATH=python python python/develop_dimensionful_walls.py` from the repository root. Thirteen new tests independently integrate the lifted kink, check dimensionful rescaling, the solved wall sandwich and first integral, thermal restoration and tiny-bias energy gaps, radiation-unit conversion, network balance, the Higgs decay channel, spectral slopes, redshift scaling and integrated spectra. All thirteen pass, together with the forty-six existing focused flavor tests. Regeneration and publication checks are recorded separately. No new full repository regression, lattice network simulation or Lean theorem is claimed.

The original release sentence is now superseded for dimensionful spectator-wall calculations: a physical-unit tension, an explicitly modeled transition scale, conditional GW spectra and a benchmark passing declared cosmological screens are obtained here. It remains correct that the repository's existing quark-flavor branch has not become cosmologically viable. If canonical source fields are F times their dimensionless coordinates and $V=F$ to the fourth times V_{hat} , tensions scale as F cubed and temperatures as F , while zero-temperature S_4 is unchanged. That rescaling cannot select a different global vacuum or control the previously large quantum displacement. Connecting this wall benchmark to viable flavor requires a justified interaction and bias that retain actual matched CP observables, a controlled effective action, thermal and relic evolution and a calibrated network. The active mechanism session retains those matching obligations.

An optional dimensionful normalization of the actual nonet receipt is also reported. Identifying its neutral canonical Higgs coordinate 0.0424264068712 with 246 GeV gives $F=5798.2756$ GeV under the common field and F -to-the-fourth potential scaling above. Its reference mass-ten mediators then sit at 57,982.756 GeV, and the lower CP-conserving competitor's energy advantage is 1.60453 times 10 to the thirteenth GeV to the fourth. The trial S_4 remains approximately 33,396,009.27. This is an explicit conditional unit assignment, not a physical mass-spectrum fit or a lifetime determination. In particular, it supplies no stable-wall tension or wall GW curve for a non-global CP-breaking branch.

The next advance closes the matrix interface needed to apply the polynomial machinery to coupled flavor physics. The new module `python/perfectpower/flavor_matrix_response.py` computes exact polynomial determinants by fraction-free Bareiss elimination, compiles supplied relaxed responses using a bordered determinant, and establishes positive matrix and operator-norm bounds through principal minors. It consumes the other flavor session's finite Gaussian matching receipt and changes none of that session's loop, pole or mechanism calculations. The driver is `python/develop_flavor_matrix_response.py` and its result is `receipts/flavor_cosmology/matrix_response.json`. Run `PYTHONPATH=python python python/develop_flavor_matrix_response.py` to regenerate it; add `--verify` to replay the persisted packet without positivity discovery. The same focused flavor test command given below now runs forty-six tests, including fifteen new matrix tests.

The first application is to both published 21-coordinate derivative metrics. Let E be the symmetrized rational realization of $K(0)-I$ at reference mediator mass 10. For the finite 27-mediator completion, the exact matrix norm is strictly below 567/500000, which is 0.001134. For the universal 55-mediator completion it is strictly below 321/250000, which is 0.001284. These are bounds in every scalar direction, established by positive principal minors of both $b \ I+E$ and $b \ I-E$. They do not merely compare a few sampled eigenvectors. The retained matrices consist of exact binary fractions of the published floats, with each off-diagonal pair averaged rationally. This explicit symmetrization is the only change from the published matrix arrays. The result is exact for those rational matrices, not a reconstruction of the exact algebraic finite-group projectors.

The up/down source coupling block is bounded independently. If B is the ten-by-ten cross-sector block, positivity of b squared $I-B$ transpose B proves its largest singular value is below b . The finite completion has bound 391/1000000, or 0.000391; the universal completion has bound 83/200000, or 0.000415. Thus the same Gaussian completion that correlates source potentials also controls a complete scalar cross-sector normalization map. Forty-two principal minors per full metric and ten per cross-sector block are checked, for 104 exact positive principal-minor values across the two published completions. A scalar metric is not a quark kinetic matrix: these bounds are not fed into a CKM angle without the separate matched fermion calculation.

The envelopes transport continuously in mediator mass on the declared fixed-EFT Gaussian valley. The current Jacobian scales with M when its exchange quartics are held fixed, so the metric correction scales as $1/M$ squared. For every $M>0$ the retained matrix norm is therefore below $(10/M)$ squared times the corresponding reference bound, and its cross-sector norm has the same scaling. Write $\epsilon=(10/M)$ squared b . If $\epsilon<1$, every eigenvalue of the retained

metric K_M lies between $1-\epsilon$ and $1+\epsilon$. In particular it is positive for M squared greater than 0.1134 in the finite case and 0.1284 in the universal case. These are conservative sufficient conditions for the rounded matrix, not physical lower mediator-mass limits. The exact Gaussian construction already has a positive induced metric from its Gram form; it is unnecessary to identify tiny roundoff directions of a retained array with physical instabilities.

For any separately supplied positive source Hessian H , that normalization envelope implies $\lambda_i(H)/(1+\epsilon) \leq \lambda_i(H, K_M) \leq \lambda_i(H)/(1-\epsilon)$, where the middle quantity is the ordered generalized eigenvalue in $H v = \lambda K_M v$. The proof is the Rayleigh quotient bound followed by the min-max principle. The present matrix certificate does not establish the source Hessian's eigenvalues or replace the other session's exact finite-momentum pole equation. It gives a conditional all-direction normalization bound, useful before solving that equation or checking a derivative expansion. Quantum corrections, heavy thresholds and the unclassified global flavor vacuum remain separate questions.

The central reusable calculation is the relaxed response $R(x) = d(x) - g(x) \text{ transpose } H(x) \text{ inverse } o(x)$. Here g is a supplied observable gradient, o is the gradient of a supplied interaction, and d is its direct matched response. The entries can be rational polynomials in one parameter. Let $D = \det H$ and C be the determinant of the bordered matrix with rows (H, o) and $(g \text{ transpose}, 0)$. The block determinant identity gives $C = -D g \text{ transpose } H \text{ inverse } o$ wherever D is nonzero, so $R = (dD + C)/D$. This needs two determinant computations rather than a numerical inverse or all individual cofactor matrices. Rational functions of the parameter are avoided inside the polynomial elimination by exact division by the preceding pivot. Exact row pivoting handles vanishing polynomial pivots, while a degree budget and dimension budget make unsupported inputs explicit.

An important branch condition survives compilation. The denominator remains $\det H$ even if the numerator is identically zero or shares factors with it. For $H = ((1, x), (x, 1))$, $g = (1, x)$, $o = (1, 0)$ and direct response one, the compiled numerator is exactly zero while the denominator remains $1 - x$ squared. Cancellation has removed this particular response; it has not made the Hessian invertible at $x = 1$. The implementation refuses to certify that matrix as stable on the whole halfline. In flavor applications, a reduced Hessian must first remove gauge directions, and a supplied stationary background must already satisfy the physical matching hypotheses. An inverse-Hessian calculation cannot supply either fact by itself.

Matrix stability over a continuous interval is implemented as well. A symmetric polynomial H is positive definite if each leading principal minor is strictly positive. This follows from its pivoted quadratic-form factorization: the successive diagonal factors are positive ratios of those minors. To prove a scalar polynomial f positive on a closed rational interval $[a, b]$, transform it to $(1+t)$ to the degree times $f((a+b t)/(1+t))$ on $t \geq 0$, and check $f(b) > 0$ separately. The mapping covers $[a, b]$ and the separate endpoint closes the interval. The existing exact halfline Sturm operation then applies. This adds finite-interval response tests without sampling near a pole or extending a local claim beyond its stated branch. Positive-semidefinite boundary points and singular gauge-reduced limits are deliberately outside this strict interface.

The operation `protected_response` combines those matrix stability conditions with a uniform absolute response bound. If β squared D squared N squared is strictly positive throughout the same domain, then $|N/D| < \beta$ there; positivity also excludes denominator zeros. A complete packet therefore contains the supplied Hessian, gradients and direct term, its bordered determinant, the exact response numerator and unreduced denominator, every positive principal minor and the response-budget polynomial. This is ready for actual matched strong-CP or CKM gradients from the other session. It makes no such gradient up: the release's applied metric bounds are physical scalar data, while its response demonstrations are explicitly identified propagator and algebraic examples.

The first exact response example reuses the other session's rational Gaussian fixture: $H = ((3, 1/3), (1/3, 2))$, three current rows $(1, 2)$, $(2, -1)$, $(1, 1)$, and squared mediator masses 11, 13, 17. Its local two-derivative Euclidean operator is $F(t) = H + t K(0)$, $t \geq 0$. Taking $g = (1, 0)$, $o = (0, 1)$, $d = 0$ gives $N = 1/3 + (48193/5909761)t$ and $D = 53/9 + (92062063/17729283)t + (6378192/5909761)t^2$. The relaxed response equals positive $3/53$ at zero momentum; the propagator's cross entry itself has the opposite sign. Exact matrix and budget polynomials prove $F(t)$ positive and $|R(t)| < 3/50$ for every nonnegative Euclidean squared momentum. This is an exact statement about the local two-derivative fixture, not a substitution of a rational kernel for the physical B_0 loop function or the finite nonlocal propagator.

The second response demonstration uses $H = ((1, t), (t, 1))$, $g = o = (1, 0)$ and $d = 0$ on $[0, 1/2]$. Its response is $-1/(1-t^2)$; the stable-interval certificate proves its magnitude below $27/20$ throughout that closed interval. A budget of one is rejected, as is extending stability through the pole at $t = 1$. These examples establish the matrix-to-response plumbing

independently of an inferred flavor prediction. In particular, first-order phase protection from a Hermitian source remains the other session's theorem; the later determinant-phase and regenerated-portal obstructions in its finite matching chapter remain untouched.

The fifteen new tests compare polynomial Bareiss determinants with independent permutation expansions, include row pivots and identically singular matrices, check explicit inverse-Hessian values at rational points, retain a canceled singular denominator, and test continuous matrix stability, both interval endpoints, interior zero touches, spectral envelopes and cross-sector Gram bounds. They also check the complete applied 21-field packet against its pinned source receipt. All forty-six focused tests pass. Regeneration and publication validation are recorded separately, with no claim of a full repository regression or a new Lean theorem. This matrix layer now makes the highest-value handoff concrete: the matching session can supply reduced Hessian and response polynomials; the polynomial layer can turn those into continuous stability and flavor-response budget statements on a declared branch.

The polynomial machinery has its highest immediate impact as a certification layer between matched flavor mechanisms and their parameter scans. An optimizer proposes a point, while exact signed polynomial remainders can establish a statement over an entire continuous parameter range. This separates numerical discovery from an exclusion or bound that can be replayed. The implemented bridge uses the repository's existing rational polynomial and Sturm core, without changing the other session's source tensors, matching, scalar poles, running or vacuum discovery. Its first physical application closes the previously open optimization question for every retained compact decay-profile family and supplies a uniform bound on the mediator kinetic correction.

The new interface is `python/perfectpower/flavor_polynomial_bounds.py`. It provides `positive_halfline`, `rational_lower_bound` and `rational_absolute_budget`, together with verifiers that check supplied signed remainders rather than repeating root discovery. The driver is `python/develop_flavor_polynomial_bounds.py`; the receipt is `receipts/flavor_cosmology/polynomial_bounds.json`. Run `PYTHONPATH=python python python/develop_flavor_polynomial_bounds.py` to rebuild it; add `--verify` for discovery-free replay of the persisted receipt. Run `PYTHONPATH=python python -m unittest discover -s python/tests -p 'test_flavor*.py'` for the focused checks. The report verifier also checks the source receipt's byte hash, binds every rational numerator and denominator to its supplied path polynomials, and checks the exact mediator factors and ratio intervals. The new tests include corrupted remainders, truncated chains, a wrong source hash, a changed mediator mass and altered polynomial coefficients.

For a real polynomial f with rational coefficients and $f(0) > 0$, construct the Sturm chain starting from f and its derivative, then successive negative Euclidean remainders. Each member is multiplied by a positive rational factor to obtain primitive integer coefficients. This preserves its signs. Sign variations at zero use each member's constant coefficient; at positive infinity they use its leading coefficient. Equal variation counts prove that f has no positive real root. Together with $f(0) > 0$ and continuity, this proves $f(x) > 0$ on the entire closed halfline. The replay checks the initial polynomial, derivative, every signed remainder, the terminal zero remainder and the two variation counts. Repeated roots are supported by the terminal gcd; a polynomial touching zero at a positive point is intentionally rejected. Positive constants are handled separately. Failed certification is inconclusive, including cases where an inequality is true only on a restricted interval. There is no automatic promotion to a general semialgebraic decision procedure or a Lean proof.

For a supplied rational observable $F(x) = N(x)/D(x)$, positivity of $N - bD$ on the entire halfline proves $F > b$ at every $x \geq 0$ with $D(x) > 0$. Importantly, D need not be positive everywhere. A feasible rational witness supplies a finite upper bound on the infimum. This is exactly the structure of the scale-reduced trial action: $N = T^2$, $D = -2U$, and $F = S_{\max}/\pi^2$. The previous numerical shape is used only as a witness and a proposal for b . The proof covers every positive-denominator component, arbitrary large L and all narrow unsampled regions. No numerical stationary-root reconstruction, finite search interval or rounded root matching enters the result.

The driver accepts a lower bound equal to the exact witness value times $(1 - 10^{-10})$. All five retained source-path certificates pass, as do the four lifted paths at mediator masses 5, 10, 20 and 100. The best retained source family's infimum lies between 33,396,009.266787387 and 33,396,009.270126987 in dimensionless action units. These displayed decimal endpoints are approximate renderings of exact rational bounds multiplied by π^2 ; the rational endpoints divided by π^2 in the receipt are authoritative. The ratio of the original endpoint family's infimum to that of the best retained family's infimum lies between 757.735824000410 and 757.735824151957. The three almost identical best endpoint images are not assigned a unique ordering when their certified intervals overlap. This result certifies a family-level variational improvement, not completeness of the symmetry-path search or the true field-space bounce.

There is also an all-shape, all-mass control statement. Let $\delta(L)$ be the exact kinetic increase at reference mass 10. The source kinetic polynomial T and δ are both proved strictly positive for every $L \geq 0$. At fixed effective quartic couplings the same Gaussian valley has $T_M = T + (10/M)^2 \delta$. Thus the lifted scale-reduced action exceeds the source action at every admissible shape, increases as the mediator mass decreases, and its infimum is nondecreasing. The mass-10 optimum-to-optimum ratio is bounded between 1.0000715282043315 and 1.0000715284043460. At masses 5, 20 and 100 the corresponding intervals are 1.0002861284656723 to 1.0002861286657296, 1.0000178817362650 to 1.0000178819362686, and 1.0000007151703810 to 1.0000007153703812. All intervals are stored as exact fractions as well as display decimals.

The stronger uniform statement uses `rational_absolute_budget`. Positivity of $b^2 D^2 - N^2$ proves $|N/D| < b$ and also excludes every denominator zero on the halfline: at a zero of D the residual could not be positive. Applied to $N = \delta$, $D = T$ and $b = 361/10000000$, it proves $\delta/T < 0.0000361$ for every shape. Consequently $1 < S_M(L)/S_{\text{source}}(L) < [1 + (10/M)^2 \cdot 0.0000361]^2$ for every $M > 0$ and every admissible L . At mass 10 the upper ratio is exactly 1.00007220130321, or an increase below 0.007220130321 percent. The same non-strict envelope bounds the ratio of the two positive infima. This controls the kinetic error uniformly on the supplied valley. It does not control transverse paths, running, loop backreaction or higher operators; in particular it does not repair the previously identified uncontrolled quantum displacement of this benchmark.

The next high-impact uses are precise interfaces for the active matching session. A relaxed first-order flavor response has a supplied direct term minus gradient Q transpose H inverse gradient O . On a one-parameter rational coefficient slice, an exact reduced Hessian and exact gradients allow adjugate/determinant elimination to express that response as N/D . The absolute-budget certificate can then bound a strong-CP response or a matched CKM response over the whole slice, and rule out poles at the same time. This requires the other session to supply the actual matched observable, canonical gauge-reduced Hessian and a justified expansion domain. A bound on a linearized response remains a bound on that response; it is not an all-orders strong-CP theorem, a symmetry explanation for tuned coefficients or a predicted CKM phase. The module already exposes the rational budget operation, but this release deliberately assigns no actual strong-CP derivative to a toy numerator.

A second use is vacuum and pole exclusion along specified coefficient slices. With an exact symmetric reduced Hessian, positivity certificates for all leading principal minors give a sufficient positive-definiteness statement by Sylvester's criterion. Supplied characteristic polynomials and Sturm variation counts can likewise exclude real eigenvalue crossings in a specified mass-squared interval. Multiple parameters, algebraic-number source tensors, gauge null directions and singular matching denominators need their own adapters; the present rational halfline implementation does not silently resolve them. These are integration targets rather than new scalar-pole or vacuum-discovery calculations, which remain owned by the other flavor session.

The release produces twelve independently replayable positivity certificates, thirteen new focused tests and eighteen existing decay tests, all executed successfully. It additionally replays the full report against the original tree-decay receipt and checks deliberate source and certificate corruption. No new Lean theorem or full repository regression is claimed. The exact statements apply to the retained binary-rational tensor realization and supplied paths, not to a reconstructed exact finite-group action or formally stationary endpoints. A lower bound on the optimized trial family is not a lower bound on the least physical bounce action and cannot establish longevity. Publication is directly on the consolidated working branch, preserving the incoming Gaussian-pole and elliptic-division advances.

The preceding decay calculation and conditional CP-wall interfaces follow with their original assumptions. Its numerical one-parameter optimization limitation is now replaced by the family-level certificates above; its endpoint, global-vacuum and quantum limitations remain.

The preceding release turns the conditional wall interface into a calculation on the published flavor action. It consumes the lower CP-conserving competitor already found by the other flavor session, rather than searching for new stationary points or changing its coefficients. At the pinned main-branch commit 22e4edf06dbdeea0f9245dbb06669ed57b7906fc, the nominated CP-breaking branch has energy -4.196243686677212, while the retained stable CP-conserving competitor has energy -4.210439239580322. The gap is 0.01419555290310992. Both are locally stationary to approximately 10^{-15} in the declared numerical action, and their reported smallest physical Hessian eigenvalues are positive. The absolute global minimum remains unclassified. This benchmark therefore calls for a metastable-decay calculation, not a claim of stable walls between global CP-breaking vacua.

The new reusable module is `python/perfectpower/flavor_decay.py`, with the driver `python/develop_flavor_cosmology.py` and eighteen focused tests in `python/tests/test_flavor_decay.py`. The scientific receipt is `receipts/flavor_cosmology/tree_decay.json`. It retains exact rational coefficients for the radial trial profiles, the input-file hashes, every chosen endpoint and the full 48-coordinate mediator-lifted path. Run `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 python python/develop_flavor_cosmology.py` from the repository root. No source couplings are retuned, no stationary points are re-minimized, and no quark matching is repeated. The earlier 104-check Node calculation remains an independent conditional interface.

The numerical source tensors, coefficients and endpoint coordinates are converted to exact binary fractions. Five exact evaluations along a source line reconstruct its quartic potential by rational Newton interpolation. This arithmetic is exact for the binary-rational realization of those arrays. It does not turn floating-point finite-group projectors into exact algebraic-number tensors, or approximate stationary coordinates into proven exact minima. Thirty-three independent ordinary floating-point evaluations per candidate check the reconstructed polynomial against the existing full potential. This distinction matters: the radial action integrals below are exact, but the physical identification with a certified bounce remains conditional on the false-vacuum and reduced-variational hypotheses.

Let $\gamma(q)$ be a polynomial path in canonically normalized real fields, with $\gamma(0)$ the supplied false vacuum and $\gamma(1)$ a lower-energy configuration. Let $W(q)=V(\gamma(q))-V(\gamma(0))$. A compact $O(4)$ trial profile has $q=1$ inside a core, a smooth shell, and $q=0$ outside. Put $u=r/R$. The core is $0 \leq u \leq L$; in the shell write $t=u-L$ between zero and one and set $q(t)=1-3t^2+2t^3$. This taper has zero derivative at both joins, so the profile is continuously differentiable. Outside $u=L+1$ the field equals the false background exactly. No exponential tail cutoff or divergent outside energy is introduced.

The complete action of this profile is

$$S_4(R,L)=2\pi^2 [T(L) R^2 + U(L) R^4],$$

where $T(L)=\text{one half times the integral from zero to one of } (L+t)^3 \text{ times the squared norm of } d\gamma(q(t))/dt$. The potential term is $U(L)=W(1)L^4/4$ plus the integral from zero to one of $(L+t)^3 W(q(t))$. Both are rational polynomials in L whenever the supplied field-path and potential coefficients are rational. T has degree at most three and U has degree at most four. The core-volume contribution ensures $U<0$ for sufficiently large L whenever $W(1)<0$. This makes a negative-action escape configuration available without pretending the Euclidean bounce is a global action minimum.

For a retained shape with $T>0$ and $U<0$, the action as a function of size starts at zero, reaches a positive maximum and then becomes negative. The maximum lies at $R_{\text{crit}}^2=-T/(2U)$, with $S_{\text{max}}/\pi^2=-T^2/(2U)$. Its derivative under radial scaling vanishes there and its scaling second derivative is negative, as expected for the escape direction. The code evaluates these quantities as exact fractions. It uses a numerical one-parameter search only to choose a rational L and then recomputes the retained action exactly. It does not certify global optimality even within the shape family.

The appropriate variational interpretation is an upper estimate for the least decay barrier, under the usual reduced-problem or mountain-pass existence assumptions and an exact metastable exterior. Every such trial profile produces a route from the false configuration to negative action; the least separating saddle cannot have a larger minimax level than that route. This is not a lower bound on the bounce action. Consequently a large trial action cannot establish a long lifetime. The familiar scale-reduced formulation originates in Coleman, Glaser and Martin, Action Minima among Solutions to a Class of Euclidean Scalar Field Equations, Communications in Mathematical Physics 58 (1978) 211-221, DOI 10.1007/BF01609421. The multifield symmetry and reduced-problem existence assumptions are treated by Blum, Honda, Sato, Takimoto and Tobioka, $O(N)$ Invariance of the Multi-Field Bounce, arXiv:1611.04570, <https://arxiv.org/abs/1611.04570>. The present radial-profile construction and its integration are an adapter to this established framework.

A significant practical reduction comes from correctly using the internal symmetry. The driver evaluates the 1080 existing family matrices on the already-published lower endpoint, together with both sector-sign transformations. This supplies 4320 symmetry-related endpoint images; it does not produce 4320 independently minimized vacua. The source coefficients remain identical. Four nearest images and the original endpoint are retained as candidate straight paths. A central group image can repeat the same real-adjoint endpoint, so the retained list is not claimed to classify distinct physical vacua. A farther endpoint might have a better potential barrier; this finite nearest-image selection is deliberately a bounded path search, not an exhaustive optimization.

With the original endpoint, the retained compact-profile trial action is approximately 2.5305352605 times 10^{10} . The best retained symmetry-related endpoint yields approximately 3.3396009270 times 10^7 , a reduction by a factor of 757.7358241. Its squared source distance is 3.210476017821027; the original endpoint's squared distance is 11.166145333203302. Its core-to-shell ratio L is 45.94751799. These are dimensionless trial actions for the specified tree potential. Their numerical size is not a cosmological lifetime, a measured decay rate, or a statement that this action is near the actual least bounce. The result shows why symmetry-aware field-path selection materially affects even a conservative decay calculation.

The mediator completion introduces a further geometric issue. The positive Gaussian square allows the scalar potential to match the source potential exactly at $S=-J(x)/M^2$, but this equality does not erase the mediator kinetic energy when x varies across a bubble. Following that trough gives the induced field-space metric

$$ds^2 = dx^T dx + dJ^T dJ/M^4.$$

The driver constructs all 27 mediator coordinates as exact quadratic polynomials along the source line, retaining the 21 source/Higgs coordinates. The complete trial bubble therefore uses 48 canonical real fields. Its potential along the path is unchanged, while its kinetic integral increases by one half the radial integral of the squared mediator derivatives. The positive-square contribution is evaluated independently at seventeen path points; it is negligible at floating-point reconstruction precision. Potential matching and kinetic matching are thus kept separate.

At any fixed path and retained shape, the completion changes the candidate action by the exact factor $[1+\Delta T(L)/T_{\text{source}}(L)]^2$. The extra kinetic integral is nonnegative, so this factor cannot decrease the action of that particular trial profile. This is not a theorem that the true UV least bounce is greater than the EFT least bounce: UV paths can leave the trough, and the least path has not been solved. The statement concerns the explicitly retained trough-following profiles. Treating Gaussian potential matching alone as sufficient for tunneling would omit this positive geometric cost.

The mass scan keeps the EFT coefficients fixed and scales the current couplings in the existing Gaussian realization proportionally to M . Then $S=-J/M^2$ scales as $1/M$, and the induced kinetic correction scales as $1/M^2$. For masses 5,10,20,100 in the stated model units, the separately shape-optimized trough-profile action ratios to the best source candidate are approximately 1.0002861286, 1.0000715283, 1.0000178818, and 1.0000007153. At $M=10$ the lifted action is approximately 3.3398398030 times 10^7 , an increase of 0.00715283 percent. This scan demonstrates decoupling within the same completed tree construction. It does not vary the mediator mass while incorrectly freezing its current normalization, and it includes no mediator-loop Coleman–Weinberg correction.

The CP interpretation also becomes sharper. Direct evaluation of the supplied generalized-CP partner reproduces the false-branch energy, and CP covariance is checked on the false point, chosen lower point and their midpoint. CP-even matching or subsequent CP-even thermal evolution cannot select a weak-CP sign merely by splitting this conjugate pair. If the CP-conserving lower basin ultimately occupies the universe, it removes the physical weak-CP order parameter instead of predicting its observed sign. Whether this basin is populated requires nucleation dynamics, initial conditions and an adequately controlled action; the local CP-breaking benchmark alone supplies none of those.

The upstream quantum calculation is a substantial limit on physical interpretation. At its fixed $\mu=1$ inputs, the linearized one-loop vacuum displacement is 15.51581387 times the tree-background norm. This signals an uncontrolled small-displacement expansion. A tree trial action cannot be presented as the quantum-corrected tunneling exponent for that benchmark. The current result deliberately imports and displays that warning alongside its own action rather than fitting counterterms or changing couplings to conceal it. A controlled quantum action is an input the mechanism session must eventually supply before physical decay claims become meaningful.

The upstream leading high-temperature minimal-action Hessian is positive, supporting asymptotic restoration within its declared expansion. That does not provide a transition temperature or an $O(3)$ nucleation action. The new calculation is zero-temperature, flat-space $O(4)$, with gravity omitted. A finite-temperature extension requires the resummed effective potential, the thermally active mediator content, possible friction and the maximum post-inflation temperature. Inferring thermal bubble rates by dividing the present $O(4)$ action by temperature would be incorrect. Likewise, reinstating a stable-wall calculation requires a candidate whose CP-breaking vacua actually lie on the global branch.

The eighteen Python tests independently integrate the compact shell with numerical quadrature, check rational interpolation, endpoint smoothness, action-scaling stationarity and the negative-action escape, reject invalid cores and

exterior energies, prove scale cancellation on exact fixtures, and check the increased kinetic cost of extra fields. Receipt tests verify the published energy ordering, exact retained action replay, full-potential sample agreement, the 48-field lift, mediator-current matching, monotone decoupling in the retained scan, and CP degeneracy. The scientific receipt replays byte for byte in a fresh process under the specified single-thread numerical environment. These are focused mathematical and action checks; no full repository regression or new Lean theorem is claimed for this push.

The continued coordination boundary is explicit. The other flavor session owns exact tensor closure, constrained mechanisms, running, quantum corrections, physical matching and vacuum discovery. This release consumes its c9010fb conclusions and changes only the cosmology module, driver, focused tests, cosmology receipt, plotting/rendering scripts and this monograph. It commits directly to the main working branch after checking the latest head. The arithmetic and Lean streams remain intact. Future cosmology calculations can reuse the same radial compiler on a new action without borrowing protection, hierarchy or vacuum claims from incompatible branches.

The advance is a reusable decay-path compiler tied to the repository's actual published action, together with its mediator-induced kinetic geometry and exact radial witnesses. It sharpens what has to be measured next: a controlled action and least-path calculation, a dimensionful normalization and prefactor, and a viable cosmological occupation history. The conditional CP-wall and constrained-bias results from the preceding release remain useful for future globally CP-breaking candidates; they are retained below with their original hypotheses.

This advance consumes the published flavor conclusions without changing their mechanisms, matching calculations, operator algebra, scalar vacua, or cumulative flavor monograph. It adds exact rational tools for a conditional cosmological question: if the chosen flavor branch really supplies degenerate physical CP-breaking global vacua, can a proposed cosmological history avoid their walls, and can a permitted explicit bias remove them without undoing specified first-order flavor protections? No dimensionful wall tension, transition temperature, gravitational-wave spectrum, or cosmologically viable repository benchmark is obtained in this release.

The implementation is `research/flavor_cosmology/cp_wall_tools.mjs`; its dependency-free test is `research/flavor_cosmology/test_cp_wall_tools.mjs`. Its examples and validation record are `receipts/flavor_cosmology/conditional_wall_tools.json`. Run `node research/flavor_cosmology/replay.mjs` to reproduce that receipt; `replay` retains the historical validation record and regenerates the scientific examples. Run `node research/flavor_cosmology/test_cp_wall_tools.mjs` from the repository root. These additions are isolated from the active scalar-loop and mechanism work. The exact mathematical tools were executed in the available V8 runtime; 104 assertions passed; the standalone replay reproduces the receipt byte for byte in V8. The workspace execution server failed to initialize, so neither the delivered Node command, Python, Lean, nor the full repository suite ran in this session. This difference is recorded rather than attributed to another session's tests.

The coordination snapshot is pinned to `a5d0f04aa6c8f666dfd9428747abbbdf3811fe50`, published at 2026-10-06 19:04 UTC. The arithmetic stream there adds elliptic division-fibre theorems, coordinate normalization, halving transport and rational-root completeness. Its predecessor `ff5d04ab56e417bbc359bfe2c11264ac566cc4d1` supplies exact elliptic witnesses and rational division fibres. The published flavor stream is `8f18f0a037d7936f7ffee9aca1a2a2959c19190e`, published at 18:36 UTC: the protected soft-mediator completion and scalar-loop closure. The user reports that a separate flavor session now owns the full source-quartic exact proof, constrained mechanisms, coupled matching and running, leading quantum and higher-operator responses, extended vacuum tests, physical parameters, and its publication. All of those are excluded from this branch's file edits. The monitoring scope is published GitHub changes and user-supplied session updates; uncommitted work and private session execution are not observable.

The existing 48-scalar completion has a positive full local Hessian, complete finite quark masses and nonzero physical weak CP in its chosen benchmark, with stated first-order mass-phase protection. Its 251-coefficient operator census and fixed-mass response do not establish a golden depth or a predicted 66-degree phase. The earlier exact field-content CP intertwiner is an anti-linear involution on the representations that pass its consistency checks, not a class-inverting operation valid for arbitrary added representations. Those are upstream conclusions in `docs/VALENTINER_SOFT_MEDIATION_AND_LOOP_CLOSURE.md` and `docs/VALENTINER_CP_AND_GLOBAL_MODES.md`, consumed at the pinned commit. Their current numerical vacuum is not promoted here to a classified global minimum.

Let C denote a consistent CP operation on the full theory and let v be a homogeneous stationary field configuration. If the scalar potential and kinetic metric obey $V(C\phi)=V(\phi)$ and the corresponding isometry relation, Cv is stationary

with the same energy and the same canonically measured fluctuation spectrum. An internal-symmetry-invariant CP-odd observable O obeys $O(g\phi)=O(\phi)$ and $O(C\phi)=-O(\phi)$. If $O(v)$ is nonzero, v and Cv cannot be in the same orbit of the internal unitary group: assuming $Cv=gv$ gives $O(v)=-O(v)$. For a quark-matched example, a nonzero Jarlskog invariant with nondegenerate spectra is such a diagnostic. It proves orbit distinction, not a count of all global minima. Distinct orbits alone do not prove disconnected vacuum components: a continuous degenerate path could join them. Isolated global minima after gauge reduction supply the additional hypothesis. A positive physical Hessian establishes local isolation; the repository's remaining global search must establish global minimality.

Counting all 1080 Valentiner matrices as independent physical vacua would therefore be wrong if an internal gauge redundancy identifies their configurations. Conversely, quotienting by ordinary internal transformations does not justify erasing spontaneous CP walls. CP reverses spatial orientation, and gauged parity/CP is not ordinary internal discrete gauge symmetry. McNamara and Reece, Reflections on Parity Breaking, arXiv:2212.00039, analyze exactly stable walls in the gauged-parity setting. A gravity embedding and its orientation structure are not supplied by this finite flavor representation calculation. This release leaves that embedding explicit rather than treating the phrase gauged CP as a wall-removal mechanism.

There is an exact energy result independent of any numerical minimizer. In a CP-invariant quantum theory with a compatible measure, regulator and renormalization prescription, CP-even corrections preserve equality of conjugate branch energies. Thresholds, perturbative running, higher-loop terms and CP-even higher-dimensional operators cannot alone create a CP-conjugate vacuum bias. They may change the common vacuum energy, wall profile or transition order; they do not select a sign. The thermal version holds for a CP-symmetric state, including zero CP-odd chemical potentials. A baryon asymmetry, imposed CP-odd background or explicit CP-breaking spurion changes this premise and must be included as a physical input. Full quantum CP consistency for the repository's enlarged field content is not established by its restricted elementary intertwiner alone.

QCD also supplies no automatic loophole. At a CP-symmetric bare θ and in a CP-symmetric medium, its free energy obeys $E(\theta)=E(-\theta)$. Conjugate branches with opposite induced strong phases therefore have equal QCD energies. In the illustrative susceptibility expansion $E(\theta)=\chi \theta^2/2$, adding an independent explicit common θ_0 gives $E(\theta_0+\theta_s)-E(\theta_0-\theta_s)=2 \chi \theta_0 \theta_s$. That bias relies on the explicit input θ_0 and vanishes if θ_s is zero. It is not a mechanism that follows from protected Nelson–Barr matching. Asadi, Homiller, Lu and Reece, Chiral Nelson–Barr Models: Quality and Cosmology, arXiv:2212.03882, discuss the interaction between strong-CP quality, inflation and reheating in concrete models.

The constructive wall result begins with two supplied degenerate global minima and a continuously differentiable path $\gamma(t)$, $0 \leq t \leq 1$, in canonically normalized real fields. Write $W(t)=V(\gamma(t))-V_{\min}$ and $K(t)=\text{the Euclidean squared norm of } \gamma'(t)$. Provided W is nonnegative, minimization of the one-dimensional spatial parametrization of that fixed path gives a candidate interface tension equal to the integral of $\sqrt{2 W K}$. The true infimum of the interface tension cannot exceed the value from this path. Cauchy–Schwarz then gives the useful exact squared bound

$$\sigma_{\min}^2 \leq \sigma_{\text{path}}^2 \leq 2 \left(\int_0^1 W(t) dt \right) \left(\int_0^1 K(t) dt \right).$$

One sees the first step by introducing $s=dt/dz > 0$ in the static energy: its density integrated over t is $K s^2 + W/s$, minimized pointwise at $s=\sqrt{2W/K}$. Endpoint singularities or interior zeros are understood by limiting finite-energy profiles. Additional paths can lower the upper bound; no lower bound follows. The argument assumes the declared degenerate global minima. Metastable minima with a deeper third branch do not by themselves define a stable cosmological wall.

For rational polynomial inputs, the code computes both integrals exactly with BigInt fractions. It checks that W and its derivative vanish at both endpoints, divides W by t^2 times $(1-t)^2$, and checks nonnegative Bernstein coefficients of the residual polynomial. Nonnegative Bernstein coefficients certify nonnegativity on the unit interval; failure is deliberately inconclusive. This is not a complete polynomial positivity algorithm. The code accepts a caller-supplied W , so it does not independently verify that W is the composition of a multivariate repository potential with γ . Nor does it certify the endpoints' global minimality. Those hypotheses are written into each receipt. Algebraic endpoints and a nonconstant field metric require further exact-field or interval adapters.

For the canonical one-field benchmark $V(a)=(a^2-1)^2/4$ and $\gamma(t)=2t-1$, $W=4t^2(1-t)^2$ and $K=4$. Their integrals are $2/15$ and 4 . The exact upper bound is $\sigma^2 \leq 16/15$. The known kink has $\sigma=2$

$\sqrt{2}/3$, hence $\sigma^2=8/9$; this gives an independent strict check on the variational bound. These are dimensionless toy data. No repository tension has been assigned this value. With general positive λ and v , the same canonical quartic has tension $2 \sqrt{2} \lambda v^3/3$, illustrating why a dimensionless flavor Hessian cannot fix an astrophysical wall tension without a physical normalization and scale.

An explicit bias has a separate, precisely bounded algebraic question. Let O_i be specified CP-odd operators and c_i their small real coefficients. To first order, the relaxed branch-energy difference is $b^T c$, where $b_i=2 O_i(v)$, up to the chosen sign of which branch is subtracted. The derivative of a relaxed energy is the operator value at its unperturbed stationary point; the first-order vacuum shift does not contribute because the unperturbed potential gradient vanishes. The vacuum shift itself is $-H^{-1}$ times the summed c_i gradient O_i . A protected observable Q has first-order response equal to its direct coefficient response minus gradient $Q^T H^{-1}$ gradient O_i . Physical matching must provide that direct response and the properly gauge-reduced Hessian; the present module does not calculate either.

Suppose rows of A collect responses required to vanish to first order, for example a supplied strong-CP response and specified CKM responses. Choose an explicit positive definite coefficient metric R and a budget $c^T R c \leq \rho^2$. The mathematical problem is to maximize $|b^T c|$ subject to $A c=0$. Discard redundant rows of A by exact elimination. Set $x=R^{-1} b$ and

$$z=R^{-1} b - R^{-1} A^T (A R^{-1} A^T)^{-1} A R^{-1} b.$$

Then $A z=0$, $z^T R z=b^T z \geq 0$, and the exact maximum squared bias is ρ^2 times $b^T z$. For a nonzero result, the maximizing coefficient vector is $\rho z/\sqrt{b^T z}$. This follows by taking the R -orthogonal projection of x onto $\ker A$ and applying Cauchy–Schwarz in the R metric. The module gives the rational projected direction, exact squared support and all constraint residuals. A vanishing support is a first-order impossibility result under the supplied constraints and coefficient budget, not an all-orders theorem that a cosmology is impossible.

The exact criterion has a useful sharp interpretation. Because R is positive definite, no allowed first-order bias exists exactly when b lies in the row span of A . If a CP-odd operator distinguishes the conjugate vacua but its bias functional is fully tied to the responses being protected, forbidding those responses also forbids that bias. If b has a surviving component in the constrained subspace, an algebraic first-order direction exists. Its physical size, radiative stability, higher-order strong phase, naturalness and full thermal dynamics remain upstream model calculations. Treating a cancellation $A c=0$ as a symmetry-derived Wilson-coefficient relation would exceed this result.

The hand-worked example uses $R=\text{diag}(1,2,3)$, $b=(2,2,2)$, and the response constraint $c_1+c_2=0$. Its projected direction is $(0,0,2/3)$, so the unit-budget squared maximum is $4/3$, versus $22/3$ without the constraint. A duplicate constraint $2c_1+2c_2=0$ changes nothing. In a two-operator example with $R=I$ and $b=(2,2)$, the same response constraint removes all bias: the support is exactly zero. Twenty-four additional positive definite rational examples check that the projected direction satisfies every original constraint, that its R norm equals its bias pairing, and that adding constraints cannot enlarge the support. No rows in these examples are claimed to be the actual repository strong-CP derivative.

The thermal adapter is deliberately a transparent toy. For a single real CP-odd field with $V(a,T)=\lambda(a^2-v^2)^2/4+c T^2 a^2/2$, positive λ , c and v^2 give $T_c^2=\lambda v^2/c$. Below T_c the conjugate minima have $a^2=v^2-c T^2/\lambda$ and remain exactly energy-degenerate. The example $\lambda=2$, $v^2=9$, $c=3$ gives $T_c^2=6$. The actual 48-scalar system has independent portals and mixed thermal masses. Its critical temperature, possible symmetry nonrestoration, transition order, tunneling and nonthermal restoration are open. In a thermal restoration scenario, maximum temperature after inflation matters; the final reheating temperature alone can miss an earlier hotter epoch. Preheating and inflationary fluctuations likewise need their own calculation.

For bookkeeping only, `classicalScalingScreen` records an explicitly declared horizon-curvature estimate: $\rho_{\text{wall}}=\text{area} \cdot \sigma \cdot H$ and $H_{\text{ann}}=\Delta V/(\text{annihilation} \cdot \sigma)$, with reduced Planck mass M . Equating wall energy to $3 M^2 H^2$ gives $H_{\text{dom}}=\text{area} \cdot \sigma/(3 M^2)$. In this convention, annihilation before both a supplied deadline H_d and wall domination asks for ΔV greater than $\text{annihilation} \cdot \sigma \cdot \max(H_d, H_{\text{dom}})$. All quantities, area and annihilation factors are caller inputs. This is a phenomenological screen, not a rigorous cosmological bound or a simulation.

Combining that declared screen with the exact coefficient envelope yields a conditional test: its required ΔV squared must be below ρ squared b transpose z to have any linearized coefficient direction within the chosen budget. A certified tension upper bound can supply a conservative sufficient threshold within this screen because both threshold terms increase with σ . An upper bound that fails the test cannot establish impossibility: the true tension might be smaller. Likewise, a support zero under homogeneous first-order response constraints does not exclude quadratic-order bias, inflationary removal or a different field content. Classical estimates are not promoted to universal dynamics. Babichev and collaborators, arXiv:2504.07902, find different bias dependence in their simulated real-scalar model, reinforcing the need to calibrate a physical network before claiming a gravitational-wave prediction.

The bounded handoff to the active mechanism session is now concrete. It can supply a globally classified pair, a dimensionful canonical metric and potential, physical CP consistency for the complete content, a set of allowed CP-odd operators if explicit breaking is admitted, and their direct plus relaxed strong-CP/CKM response rows. The separate cosmology layer can then consume these inputs to compute path tension bounds, constrained bias support and thermal-history conditions without editing its loop algebra or redoing its matching. If exact CP must remain fundamental, the explicit-bias route is unavailable and the remaining cosmological questions are inflationary removal, restoration avoidance and the actual spacetime realization of CP. These are distinct from choosing a locally stable flavor frame.

The contribution is the executable rational interface between supplied flavor protection constraints and cosmological bias, together with a conservative polynomial-path tension bound and a recorded concurrency boundary. CP-domain-wall existence, their standard cosmological problem, variational wall mechanics and constrained quadratic optimization are established mathematics and physics. No claim of a new fundamental wall mechanism, a new CKM prediction or publication priority is made. References are McNamara and Reece, <https://arxiv.org/abs/2212.00039>; Asadi, Homiller, Lu and Reece, <https://arxiv.org/abs/2212.03882>; Saikawa, <https://arxiv.org/abs/1703.02576>; and Babichev et al., <https://arxiv.org/abs/2504.07902>.

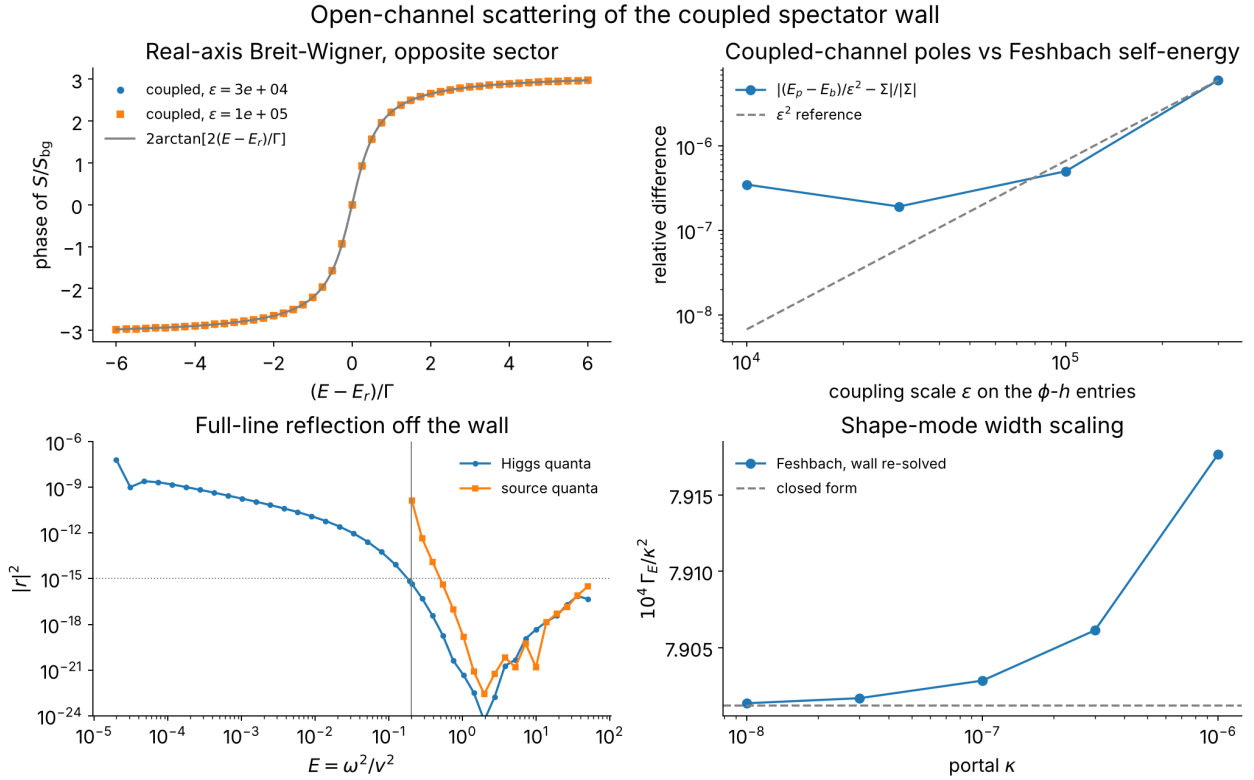
Exact thermal folds and the late-bias mechanism. The continuation now replaces a temperature scan for spinodal disappearance by an exact polynomial census. On either Higgs face the source equation is $F(u,s)=L u^3 - a(s)u - b(s)=0$, with $s=T^2/v^2$. L is $\lambda + \kappa^2/\lambda_H$ on the boundary and λ on the broken face; $a(s)$ is affine. Below the declared bias onset, $b(s)=h_0/v^3$ times $[1 - v^2 s/T_{on}^2]$, also affine. Above onset the bias is zero and this active-window polynomial must not be extrapolated. The cubic discriminant is L times $D(s)$, where $D(s)=4 a(s)^3 - 27 L b(s)^2$. Thus all possible source folds in the closed late-bias window are roots of one rational cubic per face. Sturm isolation retains every distinct root, including repeated roots and exact endpoints, and sign-at-root tests discard roots outside the physical temperature window.

For $D>0$ the unrestricted cubic has two source minima and a barrier; for $D<0$ it has one source minimum. At $D=0$ with $a=b=0$ it has the triple-root cusp. Otherwise its double source root is $u=-3b/(2a)$, the other root is $3b/a$, and $u^2=a/(3L)$ at the fold. This algebra does not by itself classify the Higgs direction. At a fold, the minimized Higgs squared amplitude is $w_{fold}(s)=\eta_0^2 + \kappa/\lambda_H - c_H s/\lambda_H - \kappa a(s)/(3 \lambda_H L)$. A boundary fold is physically Higgs-stable only for $w_{fold}<0$; a broken-face fold requires $w_{fold}>0$. Equality is a face junction and is recorded separately. This excludes unstable boundary folds from claims about a physical metastable vacuum. The Gaussian singlet curvature remains positive and the reduced source curvature is zero at the fold.

The retained 30 TeV source benchmark, its 300 TeV Gaussian mediator, the radial Higgs completion and $h_0=2.584366229533845$ times ten to the minus thirteen GeV cubed with onset 3000 GeV have positive D at both endpoints and no D roots on either face anywhere in the active late-bias interval. This is an exact exclusion for the rationalized supplied inputs, rather than a sampled-temperature observation. In particular, false-vacuum spinodal disappearance is not the removal mechanism underlying the earlier 30 MeV classical annihilation estimate. That estimate requires a dynamical network pressure balance; tunnelling, network evolution, gauge-resummed thermal corrections and the actual nonet flavor potential remain independent calculations.

Reproduce with `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 python python/develop_wall_bifurcations.py`; run `test_wall_bifurcations.py` by unittest discovery under the same `PYTHONPATH`. The receipt retains the benchmark, a physical interior fold and an unstable exact endpoint fold. Six new tests check factorization, cusp, endpoints, physical faces and independent stationary-minimum counts. All sixteen coupled-wall and thirteen dimensionful-wall tests also pass: 35 focused tests. This continuation asserts no new Lean build.

Open-channel scattering and the shape resonance of the coupled wall. The finite-box spectrum above could not decide whether the source shape mode survives once the Higgs channel is open. This continuation solves the actual scattering problem. Linear fluctuations obey $-\psi'' + H(\rho)\psi = E\psi$ with $E = \omega^2/v^2$ and H the exact three-field Hessian along the wall. Write $D = \text{diag}(-1, 1, 1)$ on (ϕ, S, η) . The composition of D with $\rho \rightarrow -\rho$ is a symmetry, because the source profile is odd and the mediator and radial Higgs profiles are even. The translation sector ($P = -1$) has ϕ Neumann and S, η Dirichlet at the center; the opposite sector ($P = +1$) has ϕ Dirichlet and S, η Neumann. The asymptotic eigenchannels are the eigenvectors of the exact vacuum Hessian, with thresholds 1.74824 times ten to the minus five (the 125.436 GeV Higgs), 0.1999992 (the 13416.38 GeV source) and 100.0004 (the 300 TeV mediator). The decoupled shape state sits at $E_b = 0.14999971951$, between the first two thresholds. In the opposite sector exactly one channel, the even Higgs wave, is therefore open, and the shape mode can only be a resonance.



All three channels are retained. On $0 \leq \rho \leq 80$, Johnson's multichannel log-derivative propagator integrates $Y = \psi' \psi$ inverse with step 0.004. Its free step is $Y \rightarrow (1 + hY)$ inverse Y ; the Hessian enters with Simpson weights and a Numerov correction at odd points, and the global error is fourth order. Beyond $\rho = 80$ the Hessian is the vacuum plus a remnant below ten to the minus eight. That remnant is dominated by the Yukawa tail of the radial Higgs, whose decay length is about 240. There the constant vacuum is propagated exactly in the eigenchannel basis, $Y \rightarrow (pS + CY)(C + (S/p)Y)$ inverse with $C = \cosh(\phi)$, $S = \sinh(\phi)$ and $p^2 = \mu - E$, and the remnant enters as Strang kicks. Open channels carry flux normalization $1/\sqrt{q}$, and the sector matrix is $S = -B/A$ for $\psi = (A e^{-iq\rho} + B e^{iq\rho})/\sqrt{q}$. The full-line matrix is $r = -(S_{plus} + S_{minus})/2$, $t = (S_{minus} - S_{plus})/2$, with right channels e_j and left channels $D e_j$. Poles are zeros of the inward Jost function $F(E) = \det(P_D X + P_N X') / \det(X' - iX)$ at $\rho = 0$, where X collects the solutions that are $e^{iq\rho}$ at the matching radius. Closed channels use $q = i \sqrt{\mu - E}$. The open Higgs channel uses the principal $\sqrt{E - \mu}$, which for $\text{Im } E < 0$ is its unphysical sheet. The outward quantity $\det(Y - iQ)$ is unsuitable: at R near 2870 its bound-state zero and pole are squeezed within $\exp(-2\kappa R)$.

The decoupled limits are reproduced. With $g = 0$ and $\kappa = 0$ the source is the $l = 2$ Poschl-Teller kink. The computed full-line source reflection probability is below ten to the minus twenty, which numerically recovers that the kink is reflectionless, and the transmission agrees with $(q + ik)(q + 2ik) / ((q - ik)(q - 2ik))$ to 9 times ten to the minus eleven for $E \leq 1$. The continuum poles appear at $E = -2.1$ times ten to the minus thirteen and at $3\kappa^2$ squared minus 2.7 times ten to the minus thirteen. With the mediator retained and $\kappa = 0$, the Higgs phases equal the free values to 9 times ten to the minus eleven. On the physical wall the translation mode is now a continuum bound state below the Higgs threshold, with pole $|E| \leq 2.1$ times ten to the minus thirteen (7.6 times ten to the minus fourteen at half step). This replaces the

box-dependent finite-element value of 0.0167 GeV squared. A quadratic finite-element calculation and the Jost zero agree on E_b to 7 times ten to the minus thirteen.

The shape resonance. Write $W(\rho)=2 \kappa u \eta$ for the ϕ - η Hessian entry. By Feshbach projection the pole is $E_b+\text{Sigma}(E_b)+O(\kappa^4)$, with $\text{Sigma}=-\langle b|W G_\eta W|b\rangle$. Here b is the normalized half-line shape state of the (ϕ, S) block, and $G_\eta(r, r')=-f_{\text{reg}}(r<) f_{\text{out}}(r>)/\text{Wronskian}$ is the outgoing Green function of the even Higgs operator, including its wall-induced potential and Yukawa tail. Numerically $\text{Sigma}=8.166 \times 10^{-18} - 3.9514 \times 10^{-18} i$. Hence $\Gamma_E=-2 \text{Im Sigma}=7.90283 \times 10^{-18}$, which equals the independent $\delta(E)$ -normalized golden rule to one part in ten to the eleven. The pole lies on the declared sheet: the Higgs channel is continued to $\text{Im } q < 0$, and the source and mediator channels are decaying. In physical units the mode has $\omega=11618.939$ GeV, width $\Gamma=3.0608 \times 10^{-13}$ GeV, lifetime 2.150×10^{-12} seconds and quality factor 3.80×10^{16} . The decay emits one radial Higgs quantum, carrying normal momentum q_v of about 11618 GeV. At this order the wall absorbs the momentum mismatch by breaking translation invariance.

The leading width has a closed form. Take the pure kink shape state $\sqrt{3\kappa} \text{sech}(\kappa \rho) \tanh(\kappa \rho)$, the coupling $2 \kappa \eta_0 \tanh(\kappa \rho)$ and the free Neumann wave $\cos(q \rho)$, with $q^2=3\kappa^2-m_h^2$ and $\nu=q/\kappa$. The Fourier transform of sech^3 is $(\pi/2)(1+\nu^2) \text{sech}(\pi \nu/2)$, which gives $\Gamma_E=3 \pi^2 \kappa^2 \eta_0^2 (\nu^2-1)^2 \text{sech}^2(\pi \nu/2)/(2\kappa q)$. For a light Higgs, ν tends to $\sqrt{3}$, and the energy decay rate becomes $\Gamma=2 \sqrt{2} \pi^2 \kappa^2 v_H^2 \text{sech}^2(\sqrt{3} \pi/2)/(\lambda_H^{3/2} v)$. The factor $\text{sech}^2(\sqrt{3} \pi/2)$, about 0.0172, is the smoothness suppression of a wall profile that is broad on the scale of the emitted wavelength. The closed form gives 7.9012×10^{-18} , 2.07 times ten to the minus four below the full calculation. A parameter scan locates the difference. It is linear in κ (1.9×10^{-5} at $\kappa=10^{-8}$ and 2.1×10^{-3} at $\kappa=10^{-6}$) and essentially unchanged at $g=0$. It is therefore the $O(\kappa)$ swelling of the radial Higgs inside the wall: $\eta(0)/\eta_0=1.06 \times 10^{-4}$, and the width scales as η^2 . The 30 TeV current changes it only at the 10^{-4} level, and λ_H between 0.10 and 0.20 changes the width by less than 3×10^{-5} . The closed form is the exact $\kappa \rightarrow 0$ limit; Γ/κ^2 runs from 7.9014 to 7.9177×10^{-4} between $\kappa=10^{-8}$ and 10^{-6} .

The physical width is about ten to the minus seventeen of E_r . That is at the floating-point floor of the Jost function: $|F|$ is 3.7×10^{-16} at the predicted pole and 1.9×10^{-11} at a 10^{-12} offset. A direct pole search at the physical coupling is therefore consistent with the result but not an independent determination, and it is not claimed as one. The independent confirmation is a coupling ladder. A factor ϵ multiplies only the ϕ - η entries of the fluctuation operator, leaving the background wall unchanged, and the poles of the full three-channel Jost function are found for $\epsilon=10^4$, 3×10^4 , 10^5 and 3×10^5 . The reduced shifts $(E_p-E_b)/\epsilon^2$ reproduce Sigma to 3.5, 1.9, 5.0 and 60 times ten to the minus seven; the last difference grows as ϵ^2 , as the next order requires. On the real axis at $\epsilon=3 \times 10^4$ and 10^5 the opposite-sector S stays unitary to 3×10^{-16} . It agrees with the decoupled background times $(E-\text{conj } E_p)/(E-E_p)$ to 10^{-4} and 10^{-5} , and its phase advances by $5.9506=4 \arctan 12$ across plus or minus six widths. A finite-box eigenvalue cannot supply this information: it moves with the box and has no width.

Background scattering. The wall is almost transparent to Higgs quanta. Their full-line reflection probability is 6.5×10^{-8} at $E=2 \times 10^{-5}$, just above threshold, 1.6×10^{-9} at 1.1×10^{-4} , 4.2×10^{-11} at 3.7×10^{-3} and 8.4×10^{-15} at 0.12. Halving κ divides it by 4.000002. Above the source threshold the heavy mediator breaks the reflectionless Poschl-Teller property. The source reflection probability is 1.4×10^{-10} at $E=0.205$ and 4×10^{-16} at 0.54, and halving g divides it by 16.0008, so it scales as g^4 . The vacuum eigenchannels rotate across the wall from e_j to $D e_j$, which flips the sign of the Higgs-source mixing angle $\theta=-8.2007 \times 10^{-9}$. At high energy the wall is therefore sudden, and the computed Higgs-to-source conversion of 2.6898×10^{-16} matches the sudden-rotation value $4 \theta^2=2.6901 \times 10^{-16}$. Reflection probabilities below about 10^{-15} are at the resolution floor, and the figure marks that line.

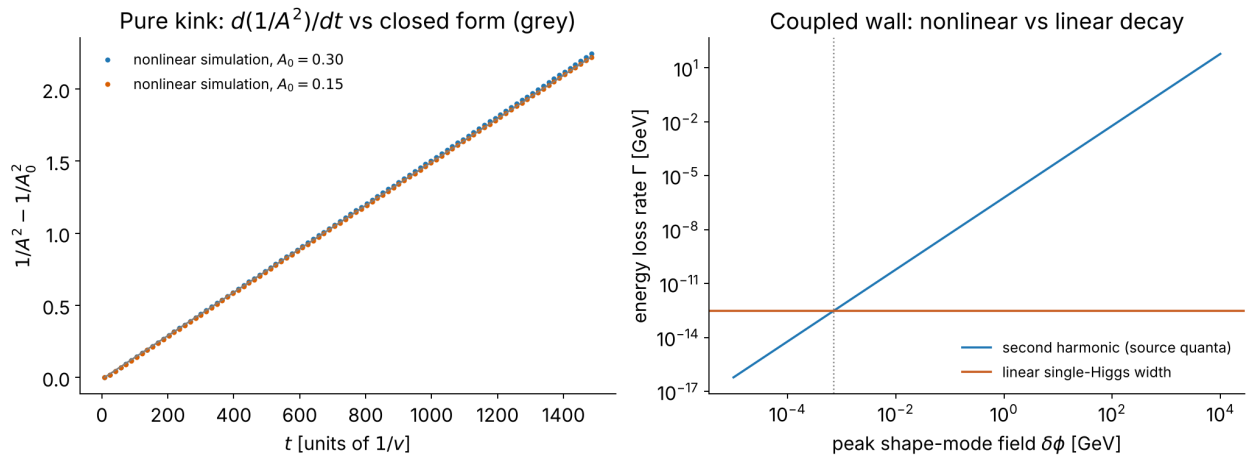
Error controls. Maximum changes of any sector phase factor are as follows: 1.2×10^{-8} under halving of the core step; 8.9×10^{-10} under halving of the tail step; 2.0×10^{-8} for

matching at 0.75 of the box; and 4.2 times ten to the minus seven at half the box, which reflects the Yukawa tail itself. Re-solving the wall in boxes of 8 and 16 Higgs lengths changes the phases by 1.0 times ten to the minus seven and 1.8 times ten to the minus nine, and the width by 5 times ten to the minus eleven relative. Halving the element spacing changes the width by 2 times ten to the minus eleven relative. Pure Johnson propagation over the whole tail with step 0.02 is off by 2.8 times ten to the minus four, which is why the exact-reference tail is used. Every one of these is a floating-point control, not an interval enclosure.

Physical reading and limits. Conditionally, for a radiation-era $H(T)$ with $g^*=106.75$, the zero-temperature single-Higgs width equals H at T of about 467 GeV. At the 60 TeV CP transition, and at the 3 TeV bias onset, linear radiation of a wall breathing excitation is therefore far slower than expansion. This comparison ignores plasma damping, thermal masses and Higgs-bath occupation, and it is not a statement about the network. The linear S-matrix also omits the amplitude-dependent nonlinear channel. Since 2ω exceeds the source threshold, a finite-amplitude shape oscillation radiates source quanta at second order. It also omits the angular Higgs and Goldstone modes, gauge fields, fermions, a continuum stability theorem and any rigorous enclosure. The new result is specific to the supplied unbiased spectator wall. It turns the open-channel question left by the finite-box spectrum into a computed resonance pole, a closed-form width, a unitary scattering matrix and the reflection laws, all with stated controls.

Reproduce with `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 python python/develop_wall_scattering.py` (about four minutes), then `python scripts/plot_wall_scattering.py`; run `test_wall_scattering.py` by unittest discovery. Eleven new tests cover: fourth-order Johnson convergence against an independent ODE; the vectorized Hessian against the published pointwise one; exactness and an overflow guard for the reference tail; Poschl-Teller transmission and poles; FEM-Jost agreement; Green function, golden rule and closed form; the coupling ladder; the real-axis Breit-Wigner; and two-channel unitarity with g to the fourth reflection scaling. The 6 bifurcation, 16 coupled-wall and 13 dimensionful-wall tests also pass. No Lean build is asserted.

Nonlinear radiation of the shape mode. The linear width above applies only to vanishingly small oscillations. Because $2\omega^2=0.6$ exceeds the source threshold 0.2, a finite-amplitude shape oscillation radiates source quanta at second order. Expand the fluctuation equation to quadratic order: $\eta_{tt}+H\eta+(1/2)V_3[\eta,\eta]=0$. Take $\eta_1=A\psi\cos(\omega t)$, with ψ the full-line normalized shape state. The quadratic term is $(A^2/4)J(1+\cos 2\omega t)$, where $J=V_3[\psi,\psi]$. Its static part only deforms the wall. The second harmonic $X e^{-2i\omega t}$ solves $(H-4\omega^2)X=-(A^2/4)J$, with outgoing or decaying conditions in every channel. The full cubic vertex of the three-field potential is: $V_{uuu}=6\lambda u+12\alpha^2 u/\mu^2+6\kappa\beta u$, $V_{uuy}=2\alpha$, $V_{uuh}=2\kappa\eta$, $V_{uhh}=2\kappa u$ and $V_{hhh}=6\lambda_H\eta$, with $\beta=\kappa/\lambda_H$. It agrees with central differences of the published Hessian to 5 times ten to the minus twelve. J is odd in ϕ and even in S and η , so the radiation lies in the opposite sector. The sector is solved with quadratic elements on $[0,150]$ in all three fields. The condition $\eta=iQ\eta$ at the outer radius, with $Q=V\text{diag}(q)V^T$, is exact for the constant vacuum. Each open channel carries time-averaged power $\Omega q|B|^2/2$ per side. The full-line loss is $P_1 A$ to the fourth, and the oscillation energy $\omega^2 A^2/2$ decays at $\Gamma_{NL}=2 P_1 A^2/\omega^2$.



On the supplied wall, $P_1=1.12907$ times ten to the minus four. It changes by 3 times ten to the minus ten under halving of the element spacing, by 4 times ten to the minus twelve when the radius grows to 220, and by 1 times ten to the minus ten when the shape-state mesh is refined. Essentially all of the power leaves in the source channel at normal momentum

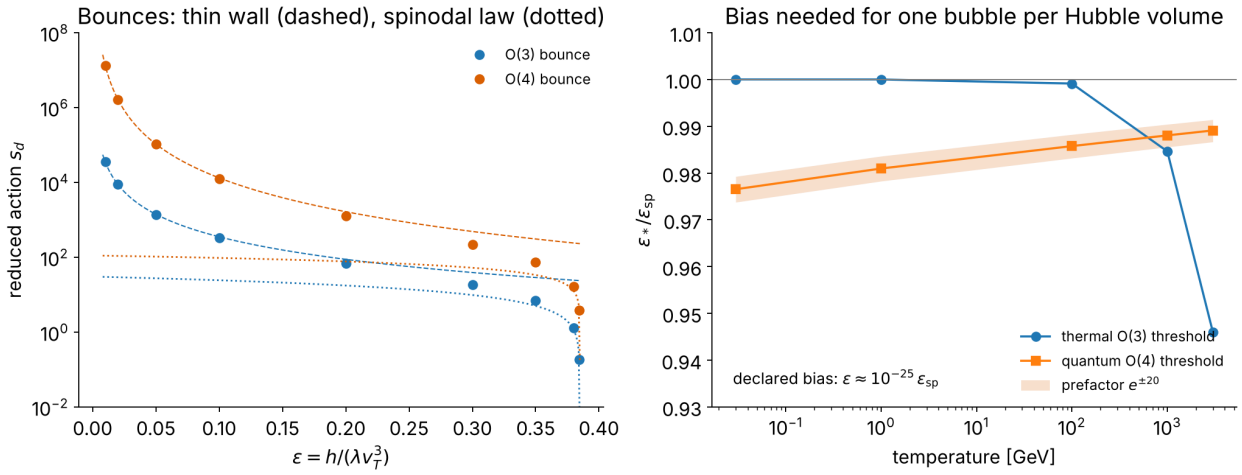
$\sqrt{0.4}$ v. The Higgs channel carries 9.8 times ten to the minus twenty-one, a share that scales as κ squared (ratio 4.0006 at double portal). The mediator and portal change P1 by -2.3 times ten to the minus five relative to the pure kink.

For the pure kink the leading loss has an exact closed form. Write $x=k\rho$. The source is $-(9\lambda k/4)\tanh^3\text{sech}$ squared, and the second harmonic has $q=2\sqrt{2}$ in units of k . Projecting onto the exact Poschl-Teller continuum function $(3\tanh^2-1-q^2-3iq\tanh)e^{iqx}$ needs the transforms of $\tanh^a\text{sech}^{2j}$, which follow from the integral of $\text{sech}^2 e^{i\nu x} = \pi\nu/\sinh(\pi\nu/2)$ and its recursions. The overlap integral is exactly $-12\pi i/\sinh(\sqrt{2}\pi)$, verified symbolically. Hence $P1=(27\sqrt{6}/32)\pi^2\lambda^2/\sinh^2(\sqrt{2}\pi)$ and $\Gamma_{NL}=(9\sqrt{6}/8)\pi^2\lambda A^2/\sinh^2(\sqrt{2}\pi)$. The finite-element value for the pure kink agrees to 2 times ten to the minus ten. The mechanism, second-harmonic radiation of the kink shape mode, is the classical one identified by Manton and Merabet (Nonlinearity 10 (1997) 3). The closed form and its use on the coupled wall are derived and checked here, without a claim of priority for the pure-kink statement. As an independent nonperturbative check, a leapfrog simulation of the full nonlinear kink with a damping sponge measures $d(1/A^2)/dt$. The ratio to the prediction is 1.0116 at $A_0=0.3$ and 1.0031 at $A_0=0.15$. The difference scales as A^2 , and extrapolation gives 1.0003.

Physical consequence. In GeV, $\Gamma_{NL}=45.163\text{ GeV} \times (\Delta\phi_{\text{peak}}/8687.2\text{ GeV})^2$, where $\Delta\phi_{\text{peak}}$ is the peak source displacement of the oscillation. The amplitude obeys $1/A^2=1/A_0^2+(2P1/\omega^2)t$, which is a power-law $t^{-1/2}$ decay rather than exponential. The nonlinear rate equals the linear single-Higgs width of 3.06 times ten to the minus thirteen GeV at $A_c=8.23$ times ten to the minus eight, a peak field of only 0.72 MeV. For any larger excitation of this 30 TeV wall, second-harmonic emission of source quanta dominates. Starting from any amplitude well above A_c , the power-law phase reaches A_c after one linear lifetime, 2.15 times ten to the minus twelve seconds; the exponential Higgs decay takes over only after that. A 1 GeV oscillation loses energy at about 6 times ten to the minus seven GeV, and a 3 TeV oscillation at about 5.4 GeV. The calculation is leading order in A . The $O(A^4)$ frequency shift, higher harmonics, plasma damping, gauge and angular Higgs modes, and network dynamics are not included.

Reproduce with `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 python python/develop_wall_shape_radiation.py` (about one minute, after `develop_wall_scattering.py`), then `python scripts/plot_wall_shape_radiation.py`. Five tests in `test_wall_shape_radiation.py` check: the vertex against the Hessian; the closed form against finite elements; convergence and the channel split; the crossover; and a shorter nonlinear simulation.

Finite-temperature transition and nucleation of the biased vacua. Two thermal questions remain open for the supplied model: whether the CP transition proceeds by bubble nucleation, and whether the late bias can remove the false vacuum by nucleation instead of network pressure. This section answers both. The declared thermal coefficient $c=0.025000033333$ equals the one-loop Debye value $\lambda/4+\kappa/3$ of the field content to every printed digit. Because ϕ is a gauge singlet, its one-loop potential is gauge independent; gauge fields enter only through Higgs loops suppressed by $\kappa=10^{-7}$. The heavy mediator has $M/T_c=4.94$. Its thermal loop shifts the ϕ mass squared through mixing, $M_S^2(\phi)=M^2+4g^2\phi^2/M^2$, by 1.7 times ten to the minus five of cT^2 squared, computed with the exact Bose integral. The resummed potential along ϕ is therefore the Arnold-Espinosa form $(cT^2-\lambda v^2)\phi^2/2+\lambda\phi^4/4-(T/12\pi)(cT^2-\lambda v^2+3\lambda\phi^2)^{3/2}$. Its only barrier comes from the self-loop of ϕ itself.



That barrier is not controlled. Read literally, the resummed potential has degenerate minima at $T_c=60768.72$ GeV, 1.28 percent above $T_0=59999.96$ GeV, with $\phi_c/T_c=0.0589$. At the broken minimum, however, the resummed mass is $m/T=0.0408$ and the three-dimensional loop parameter $\lambda T/(8 \pi m)$ equals 0.097431. That number is the same, to twelve digits, for $\lambda=0.01$, 0.1 and 1.0. In the cubic approximation $m/T=3 \lambda/(2 \pi)$ exactly, so the parameter is a pure number independent of the coupling. The would-be first-order signal is therefore never parametrically perturbative. A real Z2 scalar belongs to the three-dimensional Ising universality class, where the transition is continuous. Bubble nucleation is consequently not the mechanism of the CP transition in this model: domains and walls form through the Kibble mechanism. A conditional Kibble-Zurek estimate uses $\nu=0.63$, quench time $1/H(T_c)$ with $g^*=107.75$, and the mean-field microscopic scale $1/(\sqrt{2c}T_c)$. It gives an initial correlation length of 4.6 or 0.21 GeV^{-1} for dynamic exponents $z=1$ or 2, that is 4 times 10^7 to 9 times 10^8 domains per Hubble length. These are order-of-magnitude inputs to a network, not a simulation.

Bounces of the biased vacuum. Below T_c the biased source potential $\lambda/4(\phi^2 - v_T^2)^2$ reduces exactly to $(x^2 - 1)^2/4 - \epsilon x$ with $\epsilon = h/(\lambda v_T^3)$. Two minima exist for $0 < \epsilon < \epsilon_{sp} = 2/(3 \sqrt{3})$. The thermal and quantum actions are $S_3/T = (v_T/(\sqrt{\lambda} T)) s_3(\epsilon)$ and $S_4 = s_4(\epsilon)/\lambda$. The O(3) and O(4) bounces are found by overshoot-undershoot shooting. The shooting starts from the exact linearized solution about the true vacuum: $\sinh(z)/z$ for $d=3$ and $2 I_1(z)/z$ for $d=4$, evaluated in log space so that exponentially thin walls remain reachable. Every bounce satisfies the virial identity $(d-2)T + dU = 0$ to 4 times ten to the minus eight or better. Against the thin-wall formulas $16 \pi \sigma^3/(3 \Delta V^2)$ and $27 \pi^2 \sigma^4/(2 \Delta V^3)$, with $\sigma = 2 \sqrt{2/3}$ and the exact ΔV , the ratios are 0.99944 and 0.99936 at $\epsilon=0.01$ and 0.99778 and 0.99743 at $\epsilon=0.02$, approaching one as $1-5.5 \epsilon^2$. Near the spinodal, about the merging point $x = -1/\sqrt{3}$, the potential reduces exactly to $(1/2)m^2 z^2 - (\sqrt{3}/3)z^3$ with $m^2 = 2 \times 3^{1/4} \Delta^{1/2}$, where $\Delta = \epsilon_{sp} - \epsilon$. Hence $s_d = C_d \Delta^{(6-d)/4}$ with $C_d = (2 \times 3^{1/4})^{(6-d)/2} s_{cubic}(d)/3$. An independent shooting of $w^2/2 - w^3/3$ gives $s_{cubic}=43.660$ for $d=3$ and 204.43 for $d=4$, so $C_3=62.148$ and $C_4=179.36$. The full bounces at $\Delta=10^{-4}$, 10^{-5} and 10^{-6} approach these constants with $\Delta^{1/2}$ corrections.

Three-field bounces. For $d=3$ and 4, the least-action bounce minimizes the reduced functional $(2/d)((d-2)/d)^{(d-2)/2} T^{d/2}/(-U)^{(d-2)/2}$ over configurations with negative potential functional U ; this is the characterization of Coleman, Glaser and Martin, taken here as the hypothesis for the three-component field. The three-field potential is the biased source quartic plus two nonnegative squares, and it has the same false-vacuum energy. Projecting onto ϕ lowers T and raises $-U$, so $S_1 \leq S_3$. The valley lift $S = -\alpha u^2/\mu^2$, $\eta^2 = \eta_0^2 - \beta(u^2 - 1)$ leaves U unchanged and adds a kinetic fraction δ , so $S_3 \leq S_1(1 + \delta)^{d/2}$. On every critical bounce below, δ lies between 6.4 times ten to the minus seven and 1.2 times ten to the minus six, so the three-field action is fixed by the single-field one to 1.8 times ten to the minus six.

Nucleation threshold. Require one bubble per Hubble volume and Hubble time. With $\Gamma = T^4(S_3/2 \pi T)^{3/2} e^{-S_3/T}$ or $v_T^4(S_4/2 \pi)^2 e^{-S_4}$, the unknown determinant ratio is a declared factor of $e^{\pm 20}$. The required actions are $S_3/T=188, 169, 151, 141$ and 137 at $T=0.03, 1, 100, 1000$ and 2999 GeV ($g^*=10.75$ below 0.2 GeV and 107.75 above), and $S_4=246$ to 148. The corresponding critical biases are $\epsilon^*/\epsilon_{sp}=0.977, 0.981$ and 0.986 from quantum O(4) tunnelling at 0.03, 1 and 100 GeV. Thermal O(3) nucleation is easier above about 1 TeV,

giving 0.985 at 1 TeV and 0.946 at 2999 GeV. At 30 MeV and 1 GeV the thermal requirement is within 10^{-8} of the spinodal, given by the asymptotic law. The prefactor band moves ϵ^* by about plus or minus 0.001. The declared late bias corresponds to $\epsilon^*=9.6$ times ten to the minus twenty-six; it is short of the nucleation threshold by 24.6 orders of magnitude at its strongest. Nucleation would require a bias within a few percent of erasing the false vacuum altogether. The exact fold census already excludes that spinodal over the whole active window. Neither bubbles nor spinodal disappearance therefore remove the false vacuum for the supplied inputs. The earlier pressure-balance annihilation is the only remaining mechanism, and it is a dynamical network assumption.

Scope. These results use the leading resummed high-temperature potential, single-field bounces fixed to the three-field model by the sandwich, and a declared prefactor band. The zero-temperature Coleman-Weinberg terms (relative size $\lambda/16\pi^2$, about 6 times ten to the minus four) and two-loop thermal corrections are omitted, and the universality statement is quoted, not computed. The Kibble-Zurek numbers are conditional on the stated exponents. No lattice, real-time or network simulation is claimed.

Reproduce with `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 python python/develop_wall_nucleation.py` (about five minutes), then `python scripts/plot_wall_nucleation.py`. Seven tests in `test_wall_nucleation.py` check: the virial identity and thin-wall limit; the spinodal law against the cubic reduction; monotonicity and input guards; inversion of the critical bias; the coupling independence of the loop parameter; degenerate minima at T_c ; and the lifting ratio and nucleation threshold.